

Core Lab™

RESERVOIR OPTIMIZATION

Reservoir Fluid Analysis

for

BP Exploration Operating Co Ltd

Clair 206/12a-5 DST

RFLA 20150068

DRAFT REPORT

The analyses, opinions or interpretations in this report are based on observations and material supplied by the client to whom, and for whose exclusive and confidential use, this report is made. The interpretations or opinions expressed represent the best judgement of Core Laboratories (U.K.) Limited, (all errors and omissions excepted); but Core Laboratories (U.K.) Limited and its officers and employees assume no responsibility and make no warranty or representations as to the productivity, proper operation or profitability of any oil, gas or any other mineral well formation in connection with which such report is used or relied upon.

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25th June 2015

BP Exploration Operating Company Limited
1 Wellheads Drive
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AB21 7PB

Attention: GDPR Redaction

Dear Sir,

Subject: Reservoir Fluid Analysis; Clair 206/12a-5 (12E) DST; Our file RFLA 20150068.

On 7th October 2014 Core Laboratories (U.K.) Limited received eight bottomhole samples, twelve separator gas samples, twelve separator liquid samples, twenty-nine (25 litre) atmospheric oil samples, twenty-one (11 litre) atmospheric oil samples, twenty-one (5 litre) atmospheric oil samples, sixty-three (1 litre) atmospheric oil samples, sixteen (250ml) atmospheric oil samples and two (1 litre) oil based mud samples, all taken from the subject well. The samples were subject to a programme of analysis as specified by BP Exploration Operating Company Limited. The results of all analyses are presented in the following report.

The bottomhole samples were transferred into Core Lab owned cylinders and the empty bottles returned to Schlumberger, Aberdeen on 30th March 2015. Selected separator oil and gas samples were transferred into Core Lab owned cylinders and the empty bottles returned to Schlumberger, Aberdeen on 18th June 2015. The remaining pressurised separator oil and gas samples were disposed of and the empty cylinders returned to Schlumberger, Aberdeen on 18th June 2015. Details of all sample transfers can be found in sections B and E respectively of this report.

Atmospheric oil sample 2.51 was sent to KAT Limited, Cornwall on 19th March 2015.

Atmospheric oil sample 2.52 was sent to Weatherford, Dorset on 25th May 2015.

All remaining samples will be retained at our Aberdeen facility, with cylinders on hire, until instructed otherwise by BP Exploration Operating Company Limited.

Core Laboratories (U.K.) Limited are very pleased to have been of service to BP Exploration Operating Company Limited in this work. Should any questions arise concerning the data presented in this report, or if Core Laboratories (U.K.) Limited may be of assistance in any other matter, please do not hesitate to contact us.

Yours sincerely,

Core Laboratories (U.K.) Limited

GDPR Redaction
RFL Supervisor

FS 23304

CLB Listed NYSE

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Section A - Summary of Analysis Methods and PVT Data

Summary of Analysis Methods

Bottomhole Sample Handling and Initial Validation
The opening pressures of the bottomhole samples were measured at ambient temperature.

The bottomhole samples were heated to 200°F prior to removal of any sample in order to ensure sample integrity.

Bottomhole Oil Sample Compositions - Flash/Separation Technique
The compositions of the bottomhole oil samples were determined using a flash/separation technique and gas chromatography. The liquid was flashed at 120°F and separated into gas and liquid phases. The gas was analysed by extended gas analysis technique (see below) and the liquid by temperature programmed high resolution capillary gas chromatography (see below). The two analyses were then combined mathematically to the flash gas-liquid ratio.

Stabilised Liquid Compositions
Stabilised liquid compositions were determined by temperature programmed high resolution capillary gas chromatography utilising a cold-on-column injection technique.

Gas Compositions
Gas compositions were determined using extended gas chromatography using a GPA 2286 method utilising a multi-column gas chromatograph system. A temperature programme was used in order to provide optimum detection and resolution of the heptanes plus components.

Calculation of Molar Compositions and Residual Fraction Properties
The molar compositions and residual fraction properties are calculated entities, which are derived using the measured weight % distribution and molecular weight and density properties for the carbon number groups present.

There are four main ways to calculate a molar composition:

i. balanced to a flashed oil measured molecular weight and density using pseudo component properties. This results in compositions which calculate the residual fraction properties by material balance such that the calculated whole oil molecular weight and density equals that which was measured.

ii. not balanced to a measured molecular weight using pseudo component properties. In this case, a residual fraction molecular weight and density is assigned and the calculated whole oil molecular weight and density will not necessarily equal that which was measured (un-balanced). The assigned properties are derived from systematic approach using a linear regression technique. The log mole % versus mole weight is plotted for carbon number groups C20-C35 and extrapolated to zero mole %. The mid point molecular weight between the last defined component C35 and zero composition is assigned as the C36+ molecular weight. The density corresponding to the mid point molecular weight is taken from published property tables (Katz and Firoozabadi) and assigned as the C36+ value.

iii. calculated from measured (not pseudo component) data using properties measured on distillation fractions. In this case the residual fraction properties can also be measured.

iv. calculated from pseudo component data using properties taken from published property tables (Katz and Firoozabadi).

In the context of this analysis the molar compositions of the bottomhole samples were calculated using approach i above. The flashed oil density was measured by digital densitometer in accordance with IP Method 365.

Water Content by Karl Fischer Technique (IP 356)
A weighed representative portion of residual oil from each bottomhole oil sample was injected into a titration vessel of a volumetric Karl Fischer apparatus containing a mixed solvent. The water present was titrated to a potentiometric end point using Karl Fischer reagent. The test was repeated to confirm the initial result. The bottomhole oil samples were found to contain low levels of entrained water.

Sample Selection for PVT Analysis
Following discussions with BP Exploration Operating Co Limited, bottomhole oil sample 2587-EA was selected for further PVT analysis.

Constant Composition Expansion
A portion of bottomhole oil sample 2587-EA was charged to a high pressure visual PVT cell at the relevant reservoir temperature. A constant composition expansion was performed during which the saturation pressure was determined. Pressure-volume data for the single phase and two phase fluid were also determined.

Partial constant composition expansions were also performed to determine the saturation pressure of the sample at 104°F and 140°F.

Summary of Analysis Methods

Reservoir Fluid Density
A portion of bottomhole oil sample 2587-EA was pumped into a Paar DMA HPM high pressure densitometer equipped with an evaluation unit mPDS 2000V3. The cell measures the harmonic oscillation of the built in U-tube which contains the sample. The period of oscillation is then converted to a density of the sample by the evaluation unit which had been pre-calibrated using standards of known density. The density of the reservoir fluid was measured at 5000 psig and reservoir temperature. The densities of the other single phase pressures were calculated using the relative volume data from the constant composition expansion test.

Differential Vaporisation
This was carried out in a high pressure visual cell, at reservoir temperature on the same subsample used for the CCE test from bottomhole oil sample 2587-EA. At several pressure stages, below the observed saturation pressure, the sample was stabilised. The gas evolved at each pressure stage was then pumped out of the cell and the volume, compressibility and composition were determined. The final stage was carried out at atmospheric pressure when the residual liquid was pumped out of the cell and its density and composition were determined.

Separator Test
A multi-stage separator test was carried out using a high pressure visual PVT cell. A representative portion of bottomhole oil sample 2587-EA, at a pressure above saturation pressure, was pumped into the PVT cell and stabilised at the pressure and temperature required for the first stage separation. The evolved gas was pumped out of the cell and the volume and composition were determined. This was repeated for the second stage separation. The final stage was carried out at atmospheric pressure and 60°F and the composition of the evolved gas and stocktank oil were determined.

Atmospheric Flash
An atmospheric separator test was also performed on bottomhole oil sample 2587-EA with the stocktank oil being collected at atmospheric pressure and 60°F. The composition of the flashed gas and stocktank liquid were also determined.

Viscosity
This test was carried out using a Cambridge electro-magnetic viscometer (EMV) on a portion of bottomhole oil sample 2587-EA. The instrument contains two magnetic coils inside a stainless steel body. A low mass stainless steel piston inside the measurement chamber is magnetically forced back and forth in the fluid. The time required for the piston to move a fixed distance is then very accurately related to the viscosity of the fluid in the chamber by use of calibration constants. The viscosity measurements were run from 5000 psig down to atmospheric pressure at reservoir temperature. The measurements were repeated at each pressure stage until five consistent viscosity measurements were obtained within 0.5% of each other.

Viscosity measurements in the single-phase pressure region were also performed at 140°F, 104°F and 77°F.

Sample Handling and Initial Validation - Separator Samples
The separator gas samples were heated to a temperature 20°F above sampling temperature, agitated, and allowed to stabilise. The opening pressure, air content and composition through undecanes plus, were then determined for each sample.

The bubble point pressure of each separator liquid sample at ambient temperature was measured as a quality check.

Compositional Analysis - Separator Liquid Sample
The composition of the selected separator liquid sample was determined using a flash/separation technique and gas chromatography. The liquid was flashed at 120°F and separated into gas and liquid phases. The gas was analysed by extended gas analysis technique (see page A.1, Gas Compositions) and the liquid by temperature programmed high resolution capillary gas chromatography (see page A.1, Stabilised Liquid Compositions). The two analyses were then combined mathematically to the flash gas-liquid ratio.

A portion of the flashed liquid was subject to water content by Karl Fischer technique (see page A.1, Water Content by Karl Fischer Technique).

Wellstream Calculation
Using the measured compositional data from the selected separator liquid and companion separator gas sample, the equilibrium ratios (k) were calculated and compared to theoretical values. The k value for component i is defined as:

mole % component i in separator gas per mole % component i in separator liquid.

K values were compared with theoretical values by plotting $\log_{10}k$ against boiling point of pure hydrocarbons on a linear graph. The graph of equilibrium k against temperature should approximate a straight line.

Summary of Analysis Methods

Wellstream Calculation (Continued)
The agreement between the theoretical plot and that from the laboratory data is a further check on the validity of the samples, on the reported separator conditions and the accuracy of the laboratory determined compositions.

The graphs of equilibrium k against temperature for the selected companion separator samples are presented on page F.7.

Crude Oil Inspection Data
Crude oil inspection analysis was performed on atmospheric oil sample 2.53. The analysis results and a list of the tests performed can be found in section G of this report.

Summary of PVT Data - Bottomhole Oil Sample 2587-EA

Constant Composition Expansion at 167°F		
Reservoir pressure	2877	psig
Saturation pressure (bubble-point)	1057	psig
Average single phase compressibility	6.36	x 10-6 psi from 2877 psig to 1057 psig
Density at reservoir pressure	0.8462	g cm-3
Partial Constant Composition Expansion at 140°F		
Saturation pressure (bubble-point)	1002	psig
Density at saturation pressure	0.8455	g cm-3
Partial Constant Composition Expansion at 104°F		
Saturation pressure (bubble-point)	926	psig
Density at saturation pressure	0.8582	g cm-3
Differential Vaporisation at 167°F		
Solution gas-oil ratio at saturation pressure, Rsd	147	scf/bbl of residual oil at 60°F
Relative oil volume at saturation pressure, Bod	1.101	vol/vol of residual oil at 60°F
Density at saturation pressure	0.8364	g cm-3
Reservoir Fluid Viscosity at 167°F		
Viscosity at reservoir pressure	6.108	centipoise at 2877 psig
Viscosity at saturation pressure	4.829	centipoise at 1057 psig

Separator Test Data

Pressure (psig)	Temperature (°F)	Formation Volume Factor Bofb (2)	Total Solution Gas-oil ratio (scf/bbl) Rsfb (1)	Stocktank Oil Density at 60°F (g cm-3)
1057	167	1.103	152	
80 7 0	60			0.8994 (API = 25.7°)

Atmospheric Flash Data

Pressure (psig)	Temperature (°F)	Formation Volume Factor Bofb (2)	Total Solution Gas-oil ratio (scf/bbl) Rsfb (1)	Stocktank Oil Density at 60°F (g cm-3)
1057	167	1.095	140	
0	60			0.8974 (API = 26.0°)

(1) GOR in cubic feet of gas at 14.73 psia and 60°F per barrel of stocktank liquid at 60°F.
(2) Volume of saturated fluid at 1057 psig and 167°F per volume of stocktank liquid at 60°F.

Section B - Summary of Bottomhole Samples Received and Validation Data

Reported Well and Sampling Information

Reservoir and Well Information

Field	Clair	Clair
Well	206/12a-5	206/12a-5
Reservoir fluid	Oil	Oil
Reservoir pressure	2266 psia	2265 psia
Reservoir temperature	71.4°C	71.4°C
Installation	Paul B Loyd Jnr	Paul B Loyd Jnr
Test	DST 1	DST 1
Sampling depth	1641.32 m TVDBRT	1656.61 m TVDBRT
Formation	Jurassic	Jurassic
Flow Period	Initial Flow	Initial Flow

Sampling Information

Date sampled	06/02/15	06/02/15
Time sampled	11:49	12:08
Type of samples	Bottomhole	Bottomhole
Sampling company	Schlumberger	Schlumberger
Bottomhole pressure	2266 psia	2265 psia
Bottomhole temperature	71.4°C	71.4°C
Wellhead pressure	310 psia	308 psia
Wellhead temperature	19°C	19°C
Separator pressure	184 psia	180 psia
Separator temperature	27.4°C	27°C
Choke size	26/64"	26/64"
Stable since		
H2S	0 ppm	0 ppm
CO2	0.0%	0.0%
Gas rate (std cond)	0.8 MMscf/d	1 MMscf/d
Gas gravity (air =1)	0.653	0.662
Fb factor		
Fpv factor		
Oil rate (std cond)		
Oil rate (sep cond)	1061.5 bbl/d	1009.9 bbl/d
Oil gravity 60/60	0.868	0.868
Water rate	4.4 bbl/d	0 bbl/d
BSW	0%	0%

Comment: Well and sampling information for bottomhole samples 10314-MA (1641.3 m MDBRT) and 0975-EA (1656.6 m MDBRT) as reported by Schlumberger and BP Exploration Operating Co Ltd..

Reported Well and Sampling Information

Reservoir and Well Information		
Field	Clair	Clair
Well	206/12a-5	206/12a-5
Reservoir fluid	Oil	Oil
Reservoir pressure	2596 psia	2576 psia
Reservoir temperature	74.7°C	74.7°C
Installation	Paul B Loyd Jnr	Paul B Loyd Jnr
Test	DST 1	DST 1
Sampling depth	1678.8 m TVDBRT	1727.56 m TVDBRT
Formation	Jurassic	Jurassic
Flow Period	Main Flow	Main Flow
Sampling Information		
Date sampled	09/02/15	09/02/15
Time sampled	17:57	20:20
Type of samples	Bottomhole	Bottomhole
Sampling company	Schlumberger	Schlumberger
Bottomhole pressure	2596 psia	2576 psia
Bottomhole temperature	74.7°C	74.7°C
Wellhead pressure	330 psia	329 psia
Wellhead temperature	12°C	15°C
Separator pressure	168 psia	182 psia
Separator temperature	32°C	27°C
Choke size	24/64"	24/64"
Stable since		
H2S	0 ppm	0 ppm
CO2	0.0%	1.0%
Gas rate (std cond)	0.08 MMscf/d	0.05 MMscf/d
Gas gravity (air =1)	0.64	0.65
Fb factor		
Fpv factor		
Oil rate (std cond)		
Oil rate (sep cond)	1026.6 bbl/d	1098.3 bbl/d
Oil gravity 60/60	0.868	0.866
Water rate	6.1 bbl/d	3.6 bbl/d
BSW	0%	0%

Comment: Well and sampling information for bottomhole samples 2360-EA (1678.8 m TVDBRT) and 2587-EA (1727.6 m TVDBRT) as reported by Schlumberger and BP Exploration Operating Co Ltd..

Summary of Bottomhole Samples Received and Validation Data

Bottomhole Samples

Sample Number	Cylinder Number	Sample Point (m TVDBRT)	Flow Period	Sampling:- Date	Sampling:- Time	Laboratory Opening:- Pressure (psig)	Laboratory Opening:- Temp. (°F)	Free Water Removed (cm3)
1.62	10008-MA	1641.3	Initial Flow	06/02/15	11:49	4048	60	0
1.63	10314-MA	1641.3	Initial Flow	06/02/15	11:49	4260	66	0
1.68	2554-EA	1656.6	Initial Flow	06/02/15	12:08	4087	60	0
1.69	0975-EA	1656.6	Initial Flow	06/02/15	12:08	4154	60	0
2.09	2588-EA	1678.8	Main Flow	09/02/15	17:57	4525	68	0
2.10	2360-EA	1678.8	Main Flow	09/02/15	17:57	4208	66	0
2.40	13924-QA	1727.6	Main Flow	09/02/15	20:20	4312	67	0
2.41	2587-EA	1727.6	Main Flow	09/02/15	20:20	4983	67	0

Summary of Bottomhole Sample Transfers and Sample Volumes

Cylinder Number	Core Lab Transfer Cylinder	Transfer Conditions :- Pressure (psig)	Temperature (°F)	Transfer Sample Volume (cm3)	*Sample Volume Remaining (cm3)	Date Bottomhole Sample Cylinder Returned
10008-MA	832452	5000	200	513	513	30th March 2015
10314-MA	832903	5000	200	509	509	30th March 2015
2554-EA	832453	5000	200	516	516	30th March 2015
0975-EA	832449	5000	200	515	515	30th March 2015
2588-EA	832367	5000	200	521	521	30th March 2015
2360-EA	832460	5000	200	507	507	30th March 2015
13924-QA	832882	5000	200	515	515	30th March 2015
2587-EA	832872	5000	200	514	150	30th March 2015

*Approximate sample volume remaining on 25th June 2015.

Section C - Compositional Analysis Data - Bottomhole Samples

Compositional Analysis of Bottomhole Sample 10008-MA (1641.3 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.408	0.117	0.023
N_2	Nitrogen	0.000	0.000	1.275	0.365	0.046
C_1	Methane	0.110	0.010	78.768	22.654	1.617
C_2	Ethane	0.100	0.010	6.471	1.927	0.258
C_3	Propane	0.250	0.040	4.463	1.457	0.286
iC_4	i-Butane	0.190	0.040	1.421	0.542	0.140
nC_4	n-Butane	0.370	0.070	1.781	0.773	0.200
C_5	neo-Pentane	0.000	0.000	0.024	0.007	0.002
iC_5	i-Pentane	0.500	0.120	1.211	0.706	0.227
nC_5	n-Pentane	0.470	0.110	0.856	0.584	0.187
C_6	Hexanes	1.360	0.390	1.159	1.306	0.501
	Me-Cyclo-pentane	0.520	0.140	0.228	0.433	0.162
	Benzene	0.210	0.050	0.092	0.174	0.061
	Cyclo-hexane	0.640	0.180	0.334	0.554	0.207
C_7	Heptanes	2.000	0.660	0.499	1.570	0.700
	Me-Cyclo-hexane	1.550	0.500	0.308	1.191	0.521
	Toluene	0.760	0.230	0.129	0.576	0.236
C_8	Octanes	3.100	1.160	0.284	2.294	1.166
	Ethyl-benzene	0.240	0.090	0.014	0.178	0.084
	Meta/Para-xylene	1.000	0.350	0.040	0.724	0.342
	Ortho-xylene	0.320	0.110	0.011	0.233	0.110
C_9	Nonanes	2.890	1.210	0.138	2.102	1.200
	Tri-Me-benzene	0.450	0.180	0.000	0.321	0.172
C_{10}	Decanes	3.870	1.800	0.063	2.778	1.759
C_{11}	Undecanes	4.040	1.940	0.017	2.885	1.888
C_{12}	Dodecanes	4.130	2.180	0.005	2.945	2.111
C_{13}	Tridecanes	4.610	2.640	0.001	3.293	2.565
C_{14}	Tetradecanes	4.600	2.860	0.000	3.279	2.773
C_{15}	Pentadecanes	5.060	3.410	0.000	3.610	3.309
C_{16}	Hexadecanes	4.470	3.250	0.000	3.186	3.148
C_{17}	Heptadecanes	3.920	3.040	0.000	2.798	2.951
C_{18}	Octadecanes	4.050	3.330	0.000	2.892	3.231
C_{19}	Nonadecanes	3.900	3.360	0.000	2.783	3.258
C_{20}	Eicosanes	3.360	3.020	0.000	2.394	2.930
C_{21}	Heneicosanes	3.050	2.910	0.000	2.176	2.818
C_{22}	Docosanes	2.840	2.840	0.000	2.028	2.753
C_{23}	Tricosanes	2.520	2.620	0.000	1.798	2.545
C_{24}	Tetracosanes	2.300	2.500	0.000	1.643	2.420
C_{25}	Pentacosanes	2.210	2.500	0.000	1.577	2.422
C_{26}	Hexacosanes	1.890	2.220	0.000	1.349	2.155
C_{27}	Heptacosanes	1.780	2.180	0.000	1.269	2.112
C_{28}	Octacosanes	1.650	2.090	0.000	1.176	2.030
C_{29}	Nonacosanes	1.680	2.220	0.000	1.201	2.148
C_{30}	Triacontanes	1.570	2.140	0.000	1.119	2.073
C_{31}	Hentriacontanes	1.410	1.990	0.000	1.008	1.929
C_{32}	Dotriacontanes	1.250	1.820	0.000	0.893	1.765
C_{33}	Tritriacontanes	1.140	1.710	0.000	0.811	1.654
C_{34}	Tetratriacontanes	0.910	1.410	0.000	0.652	1.369
C_{35}	Pentatriacontanes	0.750	1.190	0.000	0.534	1.154
C_{36+}	Hexatriacontanes plus	10.010	31.180	0.000	7.135	30.282
	Totals :	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means less than 0.0005.

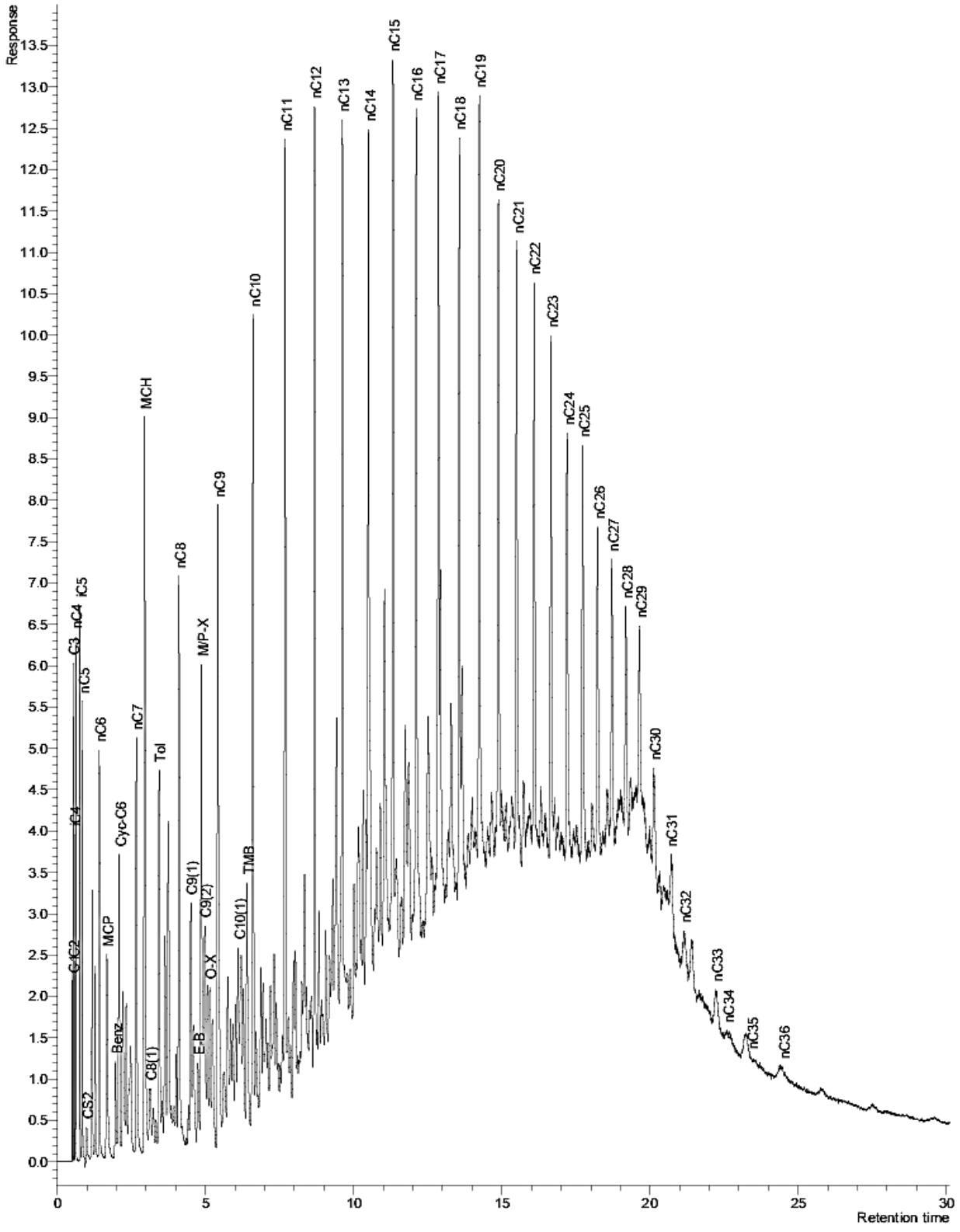
Compositional Analysis of Bottomhole Sample 10008-MA (1641.3 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.650	2.163	69.562
Molecular Weight (g mol-1)	314	99.5	312
Density at 60°F (g cm-3)	0.9035	0.7539	0.9030
C_{11+}			
Mole%	79.100	0.023	56.434
Molecular Weight (g mol-1)	358	152	358
Density at 60°F (g cm-3)	0.9177	0.7928	0.9177
C_{20+}			
Mole%	40.320		28.763
Molecular Weight (g mol-1)	504		504
Density at 60°F (g cm-3)	0.9568		0.9568
C_{36+}			
Mole %	10.010		7.135
Molecular Weight (g mol-1)	954		954
Density at 60°F (g cm-3)	1.0479		1.0479

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	305	23.6	225
Measured density at 60°F (g cm-3)	0.9004		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.819	
Entrained water content (wt%)	0.04		
Flash Gas-oil ratio (scf/bbl)			156.4

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 10008-MA (1641.3 m TVDBRT)



Compositional Analysis of Bottomhole Sample 10314-MA (1641.3 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.406	0.115	0.022
N_2	Nitrogen	0.000	0.000	1.293	0.365	0.045
C_1	Methane	0.076	0.004	79.555	22.498	1.603
C_2	Ethane	0.101	0.010	6.440	1.890	0.252
C_3	Propane	0.256	0.037	4.424	1.432	0.280
iC_4	i-Butane	0.194	0.037	1.375	0.527	0.136
nC_4	n-Butane	0.377	0.072	1.726	0.758	0.196
C_5	neo-Pentane	0.000	0.000	0.023	0.006	0.002
iC_5	i-Pentane	0.502	0.119	1.129	0.679	0.218
nC_5	n-Pentane	0.469	0.111	0.788	0.559	0.179
C_6	Hexanes	1.350	0.382	1.026	1.259	0.482
	Me-Cyclo-pentane	0.510	0.141	0.198	0.422	0.158
	Benzene	0.226	0.058	0.079	0.185	0.064
	Cyclo-hexane	0.543	0.150	0.286	0.470	0.176
C_7	Heptanes	2.009	0.661	0.420	1.561	0.695
	Me-Cyclo-hexane	1.399	0.451	0.259	1.077	0.470
	Toluene	0.757	0.229	0.100	0.572	0.234
C_8	Octanes	3.008	1.128	0.232	2.225	1.129
	Ethyl-benzene	0.232	0.081	0.011	0.170	0.080
	Meta/Para-xylene	0.967	0.337	0.033	0.703	0.332
	Ortho-xylene	0.318	0.111	0.009	0.231	0.109
C_9	Nonanes	2.840	1.196	0.112	2.071	1.179
	Tri-Me-benzene	0.418	0.165	0.000	0.300	0.160
C_{10}	Decanes	3.939	1.840	0.055	2.843	1.796
C_{11}	Undecanes	3.951	1.907	0.021	2.841	1.855
C_{12}	Dodecanes	4.111	2.173	0.000	2.952	2.111
C_{13}	Tridecanes	4.571	2.626	0.000	3.281	2.550
C_{14}	Tetradecanes	4.524	2.822	0.000	3.248	2.740
C_{15}	Pentadecanes	5.165	3.493	0.000	3.708	3.392
C_{16}	Hexadecanes	4.386	3.197	0.000	3.149	3.105
C_{17}	Heptadecanes	3.933	3.060	0.000	2.823	2.971
C_{18}	Octadecanes	4.051	3.338	0.000	2.908	3.241
C_{19}	Nonadecanes	3.800	3.281	0.000	2.728	3.186
C_{20}	Eicosanes	3.436	3.102	0.000	2.466	3.012
C_{21}	Heneicosanes	3.075	2.938	0.000	2.208	2.853
C_{22}	Docosanes	2.867	2.871	0.000	2.058	2.788
C_{23}	Tricosanes	2.542	2.654	0.000	1.825	2.577
C_{24}	Tetracosanes	2.400	2.608	0.000	1.723	2.533
C_{25}	Pentacosanes	2.101	2.380	0.000	1.508	2.311
C_{26}	Hexacosanes	1.949	2.297	0.000	1.399	2.231
C_{27}	Heptacosanes	1.774	2.178	0.000	1.273	2.115
C_{28}	Octacosanes	1.693	2.157	0.000	1.216	2.095
C_{29}	Nonacosanes	1.654	2.183	0.000	1.187	2.120
C_{30}	Triacontanes	1.633	2.230	0.000	1.172	2.165
C_{31}	Hentriacontanes	1.407	1.986	0.000	1.010	1.929
C_{32}	Dotriacontanes	1.303	1.900	0.000	0.936	1.845
C_{33}	Tritriacontanes	1.061	1.595	0.000	0.761	1.549
C_{34}	Tetratriacontanes	0.988	1.531	0.000	0.709	1.487
C_{35}	Pentatriacontanes	0.785	1.252	0.000	0.563	1.216
C_{36+}	Hexatriacontanes plus	10.349	30.921	0.000	7.430	30.026
	Totals :	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means less than 0.0005.

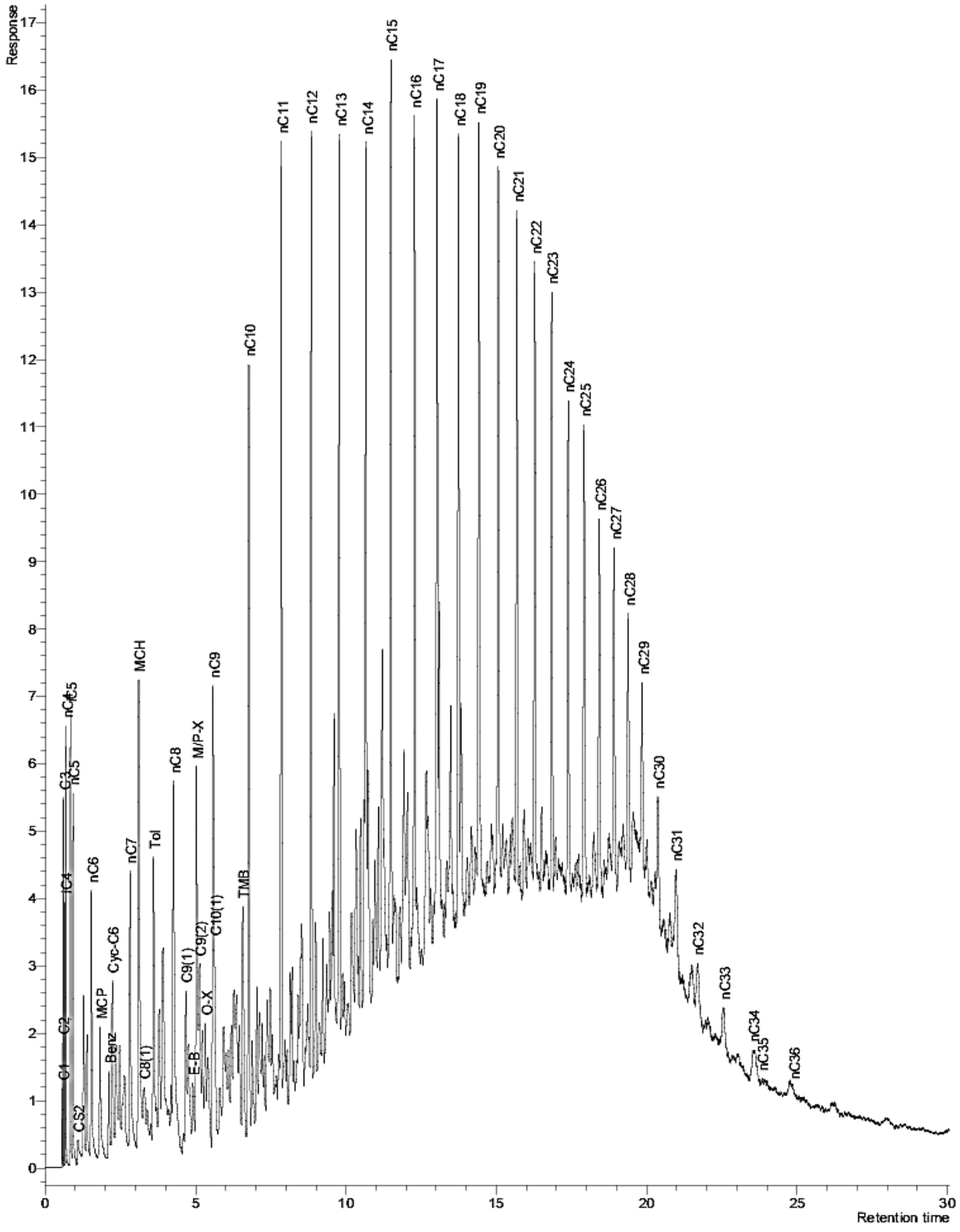
Compositional Analysis of Bottomhole Sample 10314-MA (1641.3 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.675	1.815	69.912
Molecular Weight (g mol-1)	313	99.3	311
Density at 60°F (g cm-3)	0.9032	0.7539	0.9027
C_{11+}			
Mole%	79.509	0.021	57.082
Molecular Weight (g mol-1)	355	150	355
Density at 60°F (g cm-3)	0.9171	0.7912	0.9171
C_{20+}			
Mole%	41.017		29.444
Molecular Weight (g mol-1)	496		496
Density at 60°F (g cm-3)	0.9557		0.9557
C_{36+}			
Mole %	10.349		7.430
Molecular Weight (g mol-1)	910		910
Density at 60°F (g cm-3)	1.0473		1.0473

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	305	23.1	225
Measured density at 60°F (g cm-3)	0.9001		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.801	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			153.4

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 10314-MA (1641.3 m TVDBRT)



Compositional Analysis of Bottomhole Sample 2554-EA (1656.6 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.414	0.118	0.023
N_2	Nitrogen	0.000	0.000	1.284	0.368	0.046
C_1	Methane	0.114	0.006	78.433	22.541	1.608
C_2	Ethane	0.102	0.010	6.472	1.926	0.257
C_3	Propane	0.249	0.036	4.284	1.405	0.275
iC_4	i-Butane	0.189	0.036	1.454	0.552	0.143
nC_4	n-Butane	0.368	0.070	1.842	0.790	0.204
C_5	neo-Pentane	0.000	0.000	0.026	0.007	0.002
iC_5	i-Pentane	0.504	0.119	1.284	0.727	0.233
nC_5	n-Pentane	0.474	0.112	0.911	0.600	0.192
C_6	Hexanes	1.365	0.385	1.262	1.336	0.512
	Me-Cyclo-pentane	0.516	0.142	0.249	0.439	0.164
	Benzene	0.207	0.053	0.100	0.177	0.061
	Cyclo-hexane	0.646	0.178	0.365	0.566	0.212
C_7	Heptanes	2.004	0.657	0.547	1.587	0.707
	Me-Cyclo-hexane	1.553	0.499	0.335	1.204	0.526
	Toluene	0.763	0.230	0.129	0.581	0.238
C_8	Octanes	3.106	1.161	0.311	2.306	1.171
	Ethyl-benzene	0.245	0.085	0.015	0.179	0.084
	Meta/Para-xylene	1.002	0.348	0.042	0.727	0.343
	Ortho-xylene	0.328	0.114	0.012	0.238	0.112
C_9	Nonanes	2.914	1.223	0.149	2.122	1.210
	Tri-Me-benzene	0.468	0.184	0.000	0.334	0.178
C_{10}	Decanes	3.881	1.807	0.061	2.787	1.763
C_{11}	Undecanes	4.044	1.945	0.019	2.890	1.889
C_{12}	Dodecanes	4.150	2.186	0.000	2.962	2.120
C_{13}	Tridecanes	4.605	2.637	0.000	3.286	2.557
C_{14}	Tetradecanes	4.571	2.842	0.000	3.262	2.756
C_{15}	Pentadecanes	5.130	3.458	0.000	3.661	3.353
C_{16}	Hexadecanes	4.452	3.234	0.000	3.177	3.136
C_{17}	Heptadecanes	3.956	3.068	0.000	2.823	2.975
C_{18}	Octadecanes	4.062	3.336	0.000	2.899	3.235
C_{19}	Nonadecanes	3.860	3.322	0.000	2.755	3.221
C_{20}	Eicosanes	3.364	3.027	0.000	2.401	2.935
C_{21}	Heneicosanes	3.087	2.939	0.000	2.203	2.850
C_{22}	Docosanes	2.829	2.823	0.000	2.019	2.737
C_{23}	Tricosanes	2.528	2.630	0.000	1.804	2.550
C_{24}	Tetracosanes	2.374	2.571	0.000	1.694	2.493
C_{25}	Pentacosanes	2.106	2.377	0.000	1.503	2.305
C_{26}	Hexacosanes	1.920	2.255	0.000	1.370	2.186
C_{27}	Heptacosanes	1.758	2.151	0.000	1.254	2.086
C_{28}	Octacosanes	1.663	2.111	0.000	1.187	2.047
C_{29}	Nonacosanes	1.649	2.169	0.000	1.177	2.103
C_{30}	Triacontanes	1.615	2.198	0.000	1.152	2.131
C_{31}	Hentriacontanes	1.382	1.945	0.000	0.986	1.886
C_{32}	Dotriacontanes	1.205	1.751	0.000	0.860	1.698
C_{33}	Tritriacontanes	1.138	1.706	0.000	0.812	1.654
C_{34}	Tetratriacontanes	0.962	1.486	0.000	0.687	1.441
C_{35}	Pentatriacontanes	0.783	1.245	0.000	0.559	1.207
C_{36+}	Hexatriacontanes plus	9.809	31.133	0.000	7.000	30.185
	Totals :	100.000	100.000	100.000	100.000	100.000

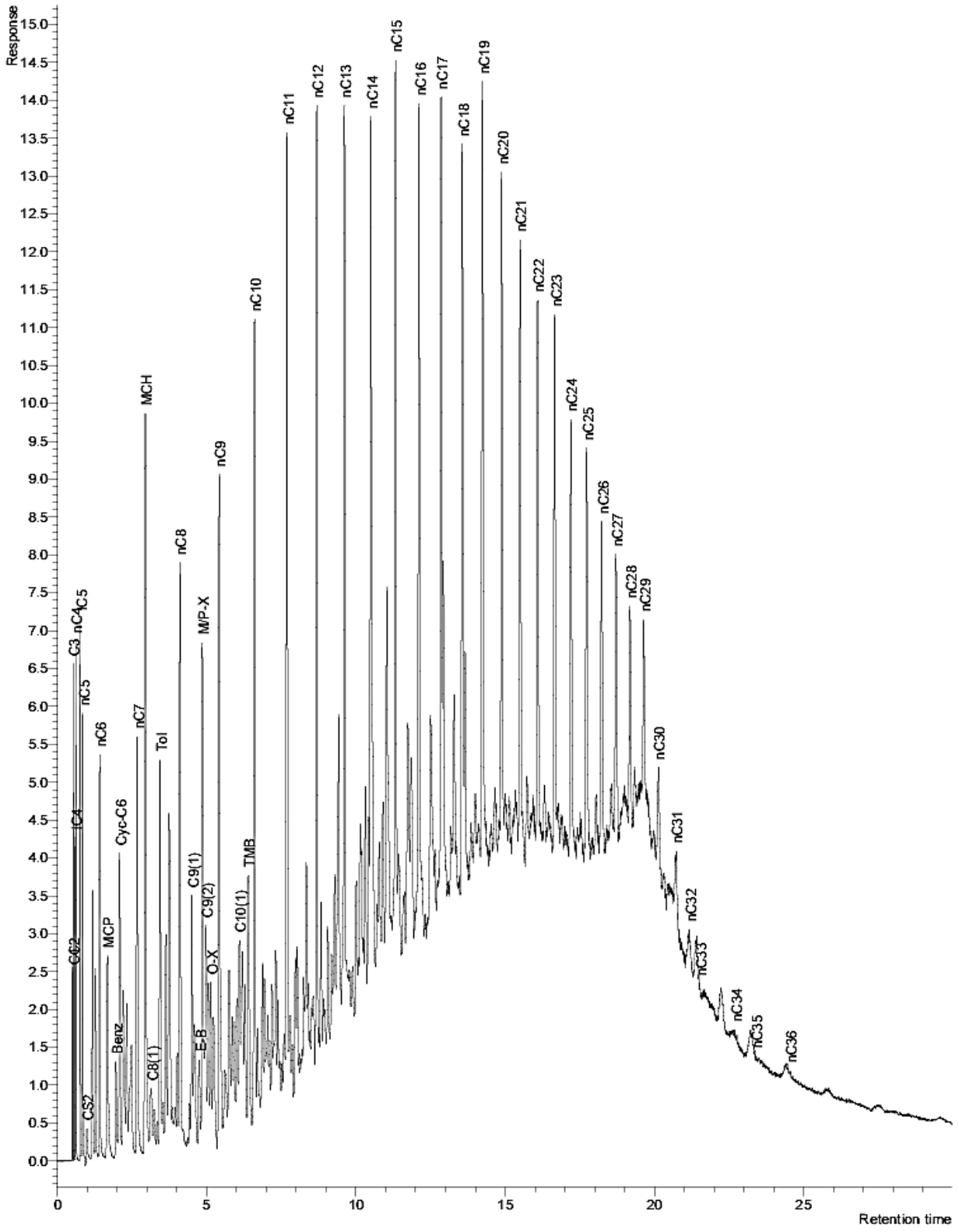
Note: 0.000 means less than 0.0005.

Compositional Analysis of Bottomhole Sample 2554-EA (1656.6 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.635	2.334	69.630
Molecular Weight (g mol-1)	314	97.1	312
Density at 60°F (g cm-3)	0.9033	0.7525	0.9027
C_{11+}			
Mole%	79.002	0.019	56.383
Molecular Weight (g mol-1)	358	150	358
Density at 60°F (g cm-3)	0.9175	0.7914	0.9175
C_{20+}			
Mole%	40.172		28.668
Molecular Weight (g mol-1)	506		506
Density at 60°F (g cm-3)	0.9566		0.9566
C_{36+}			
Mole %	9.809		7.000
Molecular Weight (g mol-1)	970		970
Density at 60°F (g cm-3)	1.0477		1.0477
Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	306	23.9	225
Measured density at 60°F (g cm-3)	0.9002		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.829	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			156.1

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 2554-EA (1656.6 m TVDBRT)



Compositional Analysis of Bottomhole Sample 0975-EA (1656.6 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.413	0.119	0.023
N_2	Nitrogen	0.000	0.000	1.286	0.370	0.046
C_1	Methane	0.134	0.007	78.761	22.760	1.624
C_2	Ethane	0.102	0.010	6.490	1.940	0.259
C_3	Propane	0.250	0.036	4.300	1.415	0.278
iC_4	i-Butane	0.190	0.036	1.463	0.556	0.144
nC_4	n-Butane	0.369	0.070	1.842	0.793	0.205
C_5	neo-Pentane	0.000	0.000	0.026	0.007	0.002
iC_5	i-Pentane	0.509	0.120	1.253	0.723	0.232
nC_5	n-Pentane	0.484	0.114	0.879	0.597	0.192
C_6	Hexanes	1.385	0.390	1.188	1.328	0.509
	Me-Cyclo-pentane	0.520	0.143	0.231	0.437	0.164
	Benzene	0.208	0.053	0.090	0.174	0.060
	Cyclo-hexane	0.647	0.178	0.333	0.557	0.209
C_7	Heptanes	2.022	0.662	0.496	1.583	0.706
	Me-Cyclo-hexane	1.568	0.503	0.302	1.204	0.526
	Toluene	0.771	0.232	0.111	0.581	0.238
C_8	Octanes	3.114	1.162	0.277	2.298	1.167
	Ethyl-benzene	0.242	0.084	0.013	0.176	0.083
	Meta/Para-xylene	1.032	0.358	0.034	0.745	0.352
	Ortho-xylene	0.329	0.114	0.010	0.237	0.112
C_9	Nonanes	2.881	1.207	0.133	2.090	1.192
	Tri-Me-benzene	0.443	0.174	0.000	0.316	0.169
C_{10}	Decanes	3.887	1.807	0.054	2.784	1.762
C_{11}	Undecanes	4.056	1.948	0.013	2.893	1.892
C_{12}	Dodecanes	4.139	2.177	0.002	2.949	2.112
C_{13}	Tridecanes	4.635	2.650	0.000	3.301	2.570
C_{14}	Tetradecanes	4.617	2.866	0.000	3.289	2.779
C_{15}	Pentadecanes	5.110	3.439	0.000	3.640	3.335
C_{16}	Hexadecanes	4.445	3.224	0.000	3.166	3.127
C_{17}	Heptadecanes	3.951	3.059	0.000	2.814	2.967
C_{18}	Octadecanes	4.080	3.346	0.000	2.906	3.245
C_{19}	Nonadecanes	3.919	3.367	0.000	2.791	3.265
C_{20}	Eicosanes	3.374	3.031	0.000	2.403	2.939
C_{21}	Heneicosanes	3.077	2.925	0.000	2.191	2.837
C_{22}	Docosanes	2.829	2.819	0.000	2.015	2.734
C_{23}	Tricosanes	2.553	2.652	0.000	1.818	2.572
C_{24}	Tetracosanes	2.331	2.521	0.000	1.660	2.445
C_{25}	Pentacosanes	2.190	2.468	0.000	1.560	2.393
C_{26}	Hexacosanes	1.859	2.180	0.000	1.324	2.114
C_{27}	Heptacosanes	1.769	2.162	0.000	1.260	2.097
C_{28}	Octacosanes	1.642	2.081	0.000	1.169	2.018
C_{29}	Nonacosanes	1.690	2.219	0.000	1.203	2.152
C_{30}	Triacontanes	1.567	2.130	0.000	1.116	2.066
C_{31}	Hentriacontanes	1.395	1.960	0.000	0.994	1.901
C_{32}	Dotriacontanes	1.276	1.851	0.000	0.909	1.795
C_{33}	Tritriacontanes	1.108	1.658	0.000	0.789	1.608
C_{34}	Tetratriacontanes	0.907	1.399	0.000	0.646	1.357
C_{35}	Pentatriacontanes	0.746	1.184	0.000	0.531	1.148
C_{36+}	Hexatriacontanes plus	9.648	31.224	0.000	6.873	30.278
	Totals :	100.000	100.000	100.000	100.000	100.000

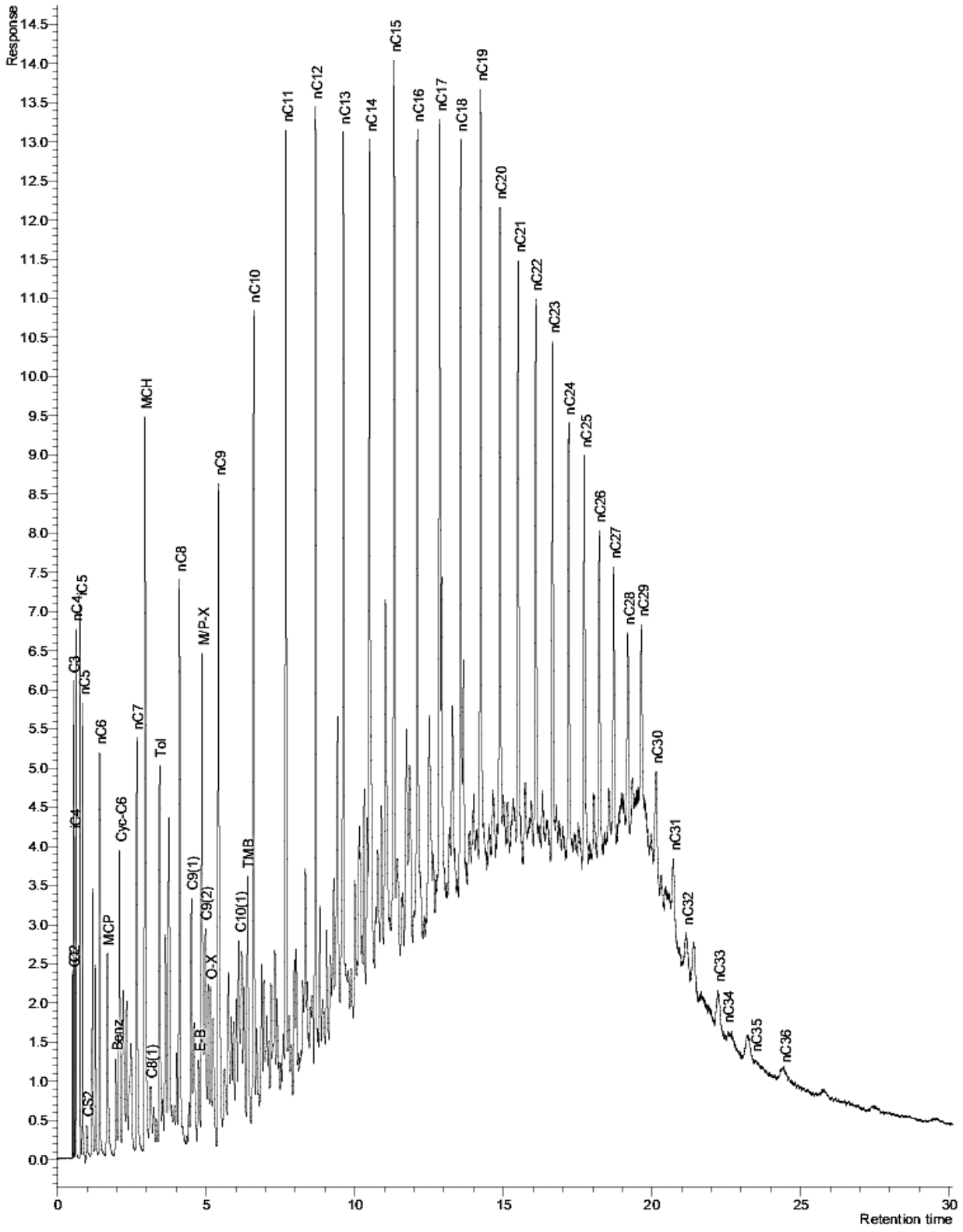
Note: 0.000 means less than 0.0005.

Compositional Analysis of Bottomhole Sample 0975-EA (1656.6 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.577	2.099	69.392
Molecular Weight (g mol-1)	314	99.0	313
Density at 60°F (g cm-3)	0.9034	0.7527	0.9029
C_{11+}			
Mole%	78.913	0.015	56.210
Molecular Weight (g mol-1)	359	149	359
Density at 60°F (g cm-3)	0.9176	0.7907	0.9176
C_{20+}			
Mole%	39.961		28.461
Molecular Weight (g mol-1)	509		509
Density at 60°F (g cm-3)	0.9569		0.9569
C_{36+}			
Mole %	9.648		6.873
Molecular Weight (g mol-1)	990		990
Density at 60°F (g cm-3)	1.0479		1.0479
Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	306	23.6	225
Measured density at 60°F (g cm-3)	0.9003		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.819	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			157.0

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 0975-EA (1656.6 m TVDBRT)



Compositional Analysis of Bottomhole Sample 2588-EA (1678.8 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.413	0.118	0.023
N_2	Nitrogen	0.000	0.000	1.283	0.366	0.046
C_1	Methane	0.114	0.006	79.093	22.614	1.621
C_2	Ethane	0.091	0.009	6.452	1.903	0.256
C_3	Propane	0.234	0.034	4.307	1.395	0.275
iC_4	i-Butane	0.178	0.034	1.468	0.545	0.142
nC_4	n-Butane	0.350	0.067	1.826	0.771	0.200
C_5	neo-Pentane	0.000	0.000	0.025	0.007	0.002
iC_5	i-Pentane	0.488	0.116	1.223	0.698	0.225
nC_5	n-Pentane	0.463	0.110	0.851	0.574	0.185
C_6	Hexanes	1.350	0.383	1.122	1.285	0.495
	Me-Cyclo-pentane	0.513	0.142	0.216	0.428	0.161
	Benzene	0.206	0.053	0.086	0.172	0.060
	Cyclo-hexane	0.635	0.176	0.310	0.543	0.204
C_7	Heptanes	1.995	0.658	0.457	1.556	0.697
	Me-Cyclo-hexane	1.534	0.496	0.277	1.176	0.516
	Toluene	0.748	0.227	0.106	0.565	0.233
C_8	Octanes	3.101	1.166	0.249	2.288	1.168
	Ethyl-benzene	0.243	0.085	0.011	0.177	0.084
	Meta/Para-xylene	1.004	0.351	0.033	0.728	0.345
	Ortho-xylene	0.320	0.112	0.010	0.232	0.110
C_9	Nonanes	2.854	1.205	0.118	2.074	1.188
	Tri-Me-benzene	0.437	0.173	0.000	0.313	0.168
C_{10}	Decanes	3.888	1.821	0.049	2.794	1.776
C_{11}	Undecanes	4.025	1.948	0.013	2.882	1.892
C_{12}	Dodecanes	4.100	2.173	0.002	2.932	2.109
C_{13}	Tridecanes	4.596	2.648	0.000	3.287	2.569
C_{14}	Tetradecanes	4.592	2.872	0.000	3.283	2.787
C_{15}	Pentadecanes	5.034	3.414	0.000	3.600	3.312
C_{16}	Hexadecanes	4.457	3.257	0.000	3.187	3.160
C_{17}	Heptadecanes	3.890	3.035	0.000	2.782	2.945
C_{18}	Octadecanes	4.036	3.335	0.000	2.886	3.236
C_{19}	Nonadecanes	3.822	3.309	0.000	2.733	3.211
C_{20}	Eicosanes	3.412	3.089	0.000	2.440	2.997
C_{21}	Heneicosanes	3.025	2.898	0.000	2.163	2.812
C_{22}	Docosanes	2.800	2.811	0.000	2.002	2.727
C_{23}	Tricosanes	2.553	2.673	0.000	1.826	2.593
C_{24}	Tetracosanes	2.301	2.507	0.000	1.645	2.432
C_{25}	Pentacosanes	2.170	2.465	0.000	1.552	2.392
C_{26}	Hexacosanes	1.872	2.212	0.000	1.338	2.146
C_{27}	Heptacosanes	1.806	2.224	0.000	1.292	2.158
C_{28}	Octacosanes	1.574	2.011	0.000	1.126	1.951
C_{29}	Nonacosanes	1.712	2.266	0.000	1.224	2.199
C_{30}	Triacontanes	1.584	2.169	0.000	1.133	2.104
C_{31}	Hentriacontanes	1.445	2.046	0.000	1.034	1.985
C_{32}	Dotriacontanes	1.234	1.804	0.000	0.883	1.750
C_{33}	Tritriacontanes	1.120	1.688	0.000	0.801	1.638
C_{34}	Tetratriacontanes	0.920	1.430	0.000	0.658	1.387
C_{35}	Pentatriacontanes	0.746	1.193	0.000	0.533	1.157
C_{36+}	Hexatriacontanes plus	10.428	31.099	0.000	7.456	30.171
	Totals :	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means less than 0.0005.

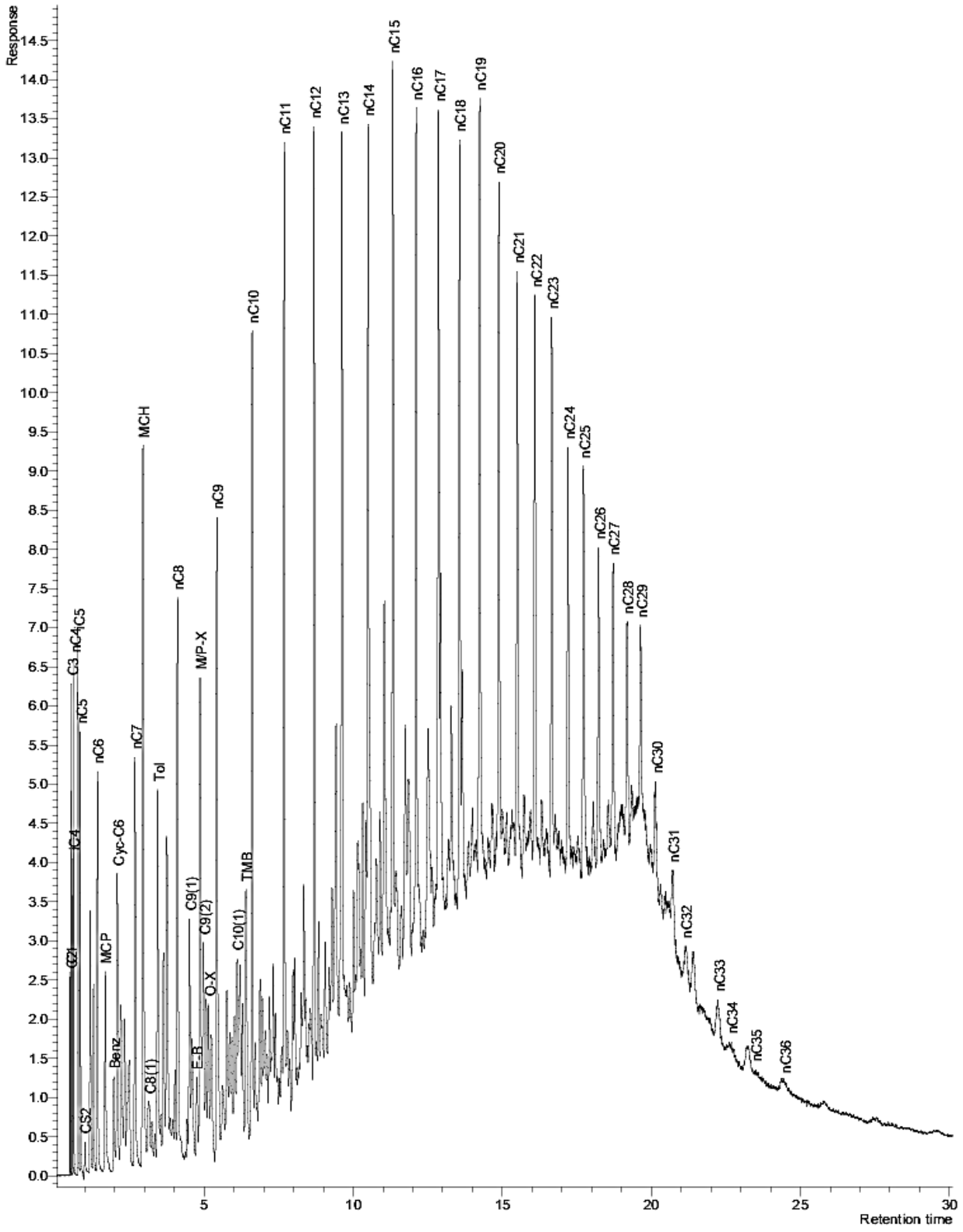
Compositional Analysis of Bottomhole Sample 2588-EA (1678.8 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.732	1.937	69.724
Molecular Weight (g mol-1)	312	98.9	310
Density at 60°F (g cm-3)	0.9030	0.7533	0.9025
C_{11+}			
Mole%	79.254	0.015	56.678
Molecular Weight (g mol-1)	355	149	355
Density at 60°F (g cm-3)	0.9172	0.7907	0.9171
C_{20+}			
Mole%	40.702		29.106
Molecular Weight (g mol-1)	497		497
Density at 60°F (g cm-3)	0.9560		0.9560
C_{36+}			
Mole %	10.428		7.456
Molecular Weight (g mol-1)	906		906
Density at 60°F (g cm-3)	1.0466		1.0466

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	304	23.4	224
Measured density at 60°F (g cm-3)	0.9000		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.811	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			156.0

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 2588-EA (1678.8 m TVDBRT)



Compositional Analysis of Bottomhole Sample 2360-EA (1678.8 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.415	0.119	0.023
N_2	Nitrogen	0.000	0.000	1.284	0.368	0.046
C_1	Methane	0.076	0.004	79.776	22.951	1.636
C_2	Ethane	0.092	0.009	6.476	1.924	0.257
C_3	Propane	0.236	0.034	4.232	1.383	0.271
iC_4	i-Butane	0.190	0.036	1.408	0.539	0.139
nC_4	n-Butane	0.364	0.069	1.744	0.760	0.196
C_5	neo-Pentane	0.000	0.000	0.024	0.007	0.002
iC_5	i-Pentane	0.501	0.118	1.121	0.679	0.218
nC_5	n-Pentane	0.467	0.110	0.773	0.555	0.178
C_6	Hexanes	1.319	0.371	0.999	1.227	0.470
	Me-Cyclo-pentane	0.517	0.142	0.191	0.424	0.158
	Benzene	0.235	0.060	0.076	0.190	0.066
	Cyclo-hexane	0.553	0.152	0.274	0.473	0.177
C_7	Heptanes	1.991	0.651	0.403	1.535	0.683
	Me-Cyclo-hexane	1.392	0.446	0.247	1.063	0.464
	Toluene	0.738	0.222	0.095	0.554	0.227
C_8	Octanes	3.128	1.166	0.225	2.295	1.165
	Ethyl-benzene	0.245	0.085	0.011	0.178	0.084
	Meta/Para-xylene	0.866	0.300	0.032	0.627	0.296
	Ortho-xylene	0.329	0.114	0.009	0.237	0.112
C_9	Nonanes	2.860	1.197	0.112	2.071	1.180
	Tri-Me-benzene	0.413	0.162	0.000	0.295	0.157
C_{10}	Decanes	4.071	1.890	0.052	2.917	1.844
C_{11}	Undecanes	3.994	1.916	0.021	2.853	1.863
C_{12}	Dodecanes	4.142	2.176	0.000	2.954	2.113
C_{13}	Tridecanes	4.553	2.600	0.000	3.246	2.524
C_{14}	Tetradecanes	4.500	2.790	0.000	3.208	2.708
C_{15}	Pentadecanes	5.129	3.448	0.000	3.657	3.347
C_{16}	Hexadecanes	4.405	3.191	0.000	3.141	3.098
C_{17}	Heptadecanes	3.942	3.049	0.000	2.811	2.960
C_{18}	Octadecanes	4.033	3.303	0.000	2.875	3.206
C_{19}	Nonadecanes	3.830	3.287	0.000	2.731	3.191
C_{20}	Eicosanes	3.348	3.004	0.000	2.387	2.916
C_{21}	Heneicosanes	3.091	2.935	0.000	2.204	2.849
C_{22}	Docosanes	2.824	2.811	0.000	2.014	2.729
C_{23}	Tricosanes	2.576	2.673	0.000	1.837	2.595
C_{24}	Tetracosanes	2.293	2.477	0.000	1.635	2.404
C_{25}	Pentacosanes	2.182	2.456	0.000	1.555	2.384
C_{26}	Hexacosanes	1.876	2.198	0.000	1.338	2.134
C_{27}	Heptacosanes	1.817	2.217	0.000	1.295	2.152
C_{28}	Octacosanes	1.633	2.068	0.000	1.165	2.007
C_{29}	Nonacosanes	1.625	2.132	0.000	1.159	2.070
C_{30}	Triacontanes	1.571	2.133	0.000	1.120	2.071
C_{31}	Hentriacontanes	1.509	2.118	0.000	1.076	2.056
C_{32}	Dotriacontanes	1.240	1.797	0.000	0.884	1.744
C_{33}	Tritriacontanes	1.085	1.621	0.000	0.773	1.574
C_{34}	Tetratriacontanes	0.935	1.440	0.000	0.667	1.398
C_{35}	Pentatriacontanes	0.759	1.204	0.000	0.541	1.169
C_{36+}	Hexatriacontanes plus	10.525	31.618	0.000	7.503	30.689
	Totals :	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means less than 0.0005.

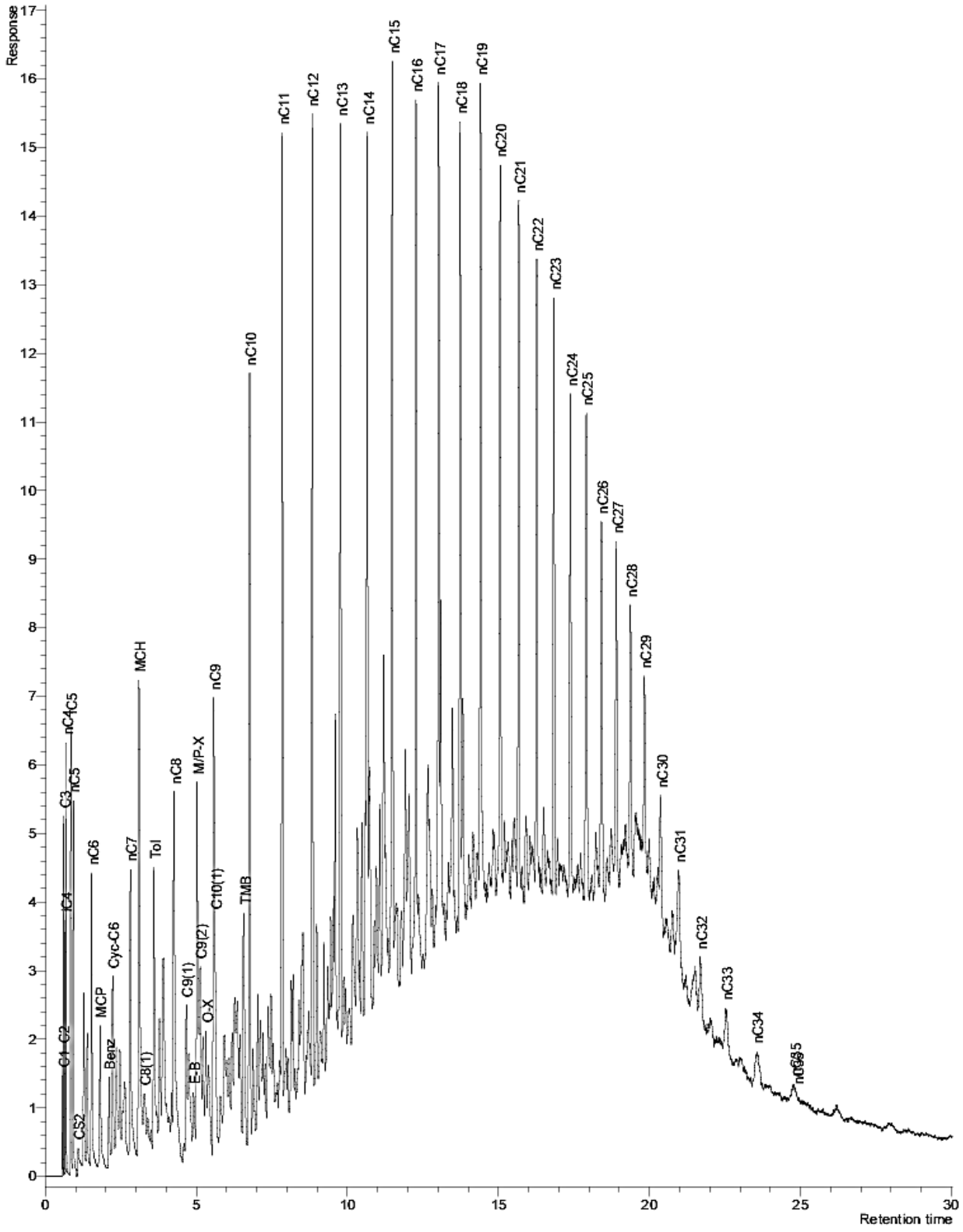
Compositional Analysis of Bottomhole Sample 2360-EA (1678.8 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.755	1.748	69.488
Molecular Weight (g mol-1)	314	99.4	313
Density at 60°F (g cm-3)	0.9029	0.7538	0.9024
C_{11+}			
Mole%	79.417	0.021	56.629
Molecular Weight (g mol-1)	358	150	358
Density at 60°F (g cm-3)	0.9170	0.7914	0.9169
C_{20+}			
Mole%	40.889		29.153
Molecular Weight (g mol-1)	501		501
Density at 60°F (g cm-3)	0.9551		0.9551
C_{36+}			
Mole %	10.525		7.503
Molecular Weight (g mol-1)	921		921
Density at 60°F (g cm-3)	1.0422		1.0422

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	306	23.0	225
Measured density at 60°F (g cm-3)	0.8999		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.797	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			156.2

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 2360-EA (1678.8 m TVDBRT)



Compositional Analysis of Bottomhole Sample 13924-QA (1727.6 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.428	0.122	0.024
N_2	Nitrogen	0.000	0.000	1.323	0.378	0.047
C_1	Methane	0.114	0.006	79.599	22.800	1.631
C_2	Ethane	0.101	0.010	6.555	1.943	0.260
C_3	Propane	0.249	0.036	4.241	1.388	0.273
iC_4	i-Butane	0.194	0.037	1.432	0.547	0.142
nC_4	n-Butane	0.383	0.073	1.775	0.780	0.202
C_5	neo-Pentane	0.000	0.000	0.025	0.007	0.002
iC_5	i-Pentane	0.536	0.127	1.149	0.711	0.229
nC_5	n-Pentane	0.503	0.119	0.787	0.584	0.188
C_6	Hexanes	1.450	0.410	1.006	1.323	0.508
	Me-Cyclo-pentane	0.532	0.147	0.191	0.435	0.163
	Benzene	0.218	0.056	0.076	0.178	0.062
	Cyclo-hexane	0.659	0.182	0.272	0.549	0.206
C_7	Heptanes	2.056	0.676	0.397	1.582	0.707
	Me-Cyclo-hexane	1.580	0.509	0.240	1.197	0.524
	Toluene	0.800	0.242	0.091	0.598	0.246
C_8	Octanes	3.113	1.167	0.213	2.286	1.164
	Ethyl-benzene	0.247	0.086	0.010	0.179	0.085
	Meta/Para-xylene	1.056	0.368	0.028	0.763	0.361
	Ortho-xylene	0.336	0.117	0.008	0.242	0.115
C_9	Nonanes	2.846	1.198	0.101	2.063	1.179
	Tri-Me-benzene	0.444	0.175	0.000	0.317	0.170
C_{10}	Decanes	3.887	1.815	0.041	2.790	1.769
C_{11}	Undecanes	4.067	1.962	0.012	2.909	1.907
C_{12}	Dodecanes	4.132	2.183	0.000	2.953	2.119
C_{13}	Tridecanes	4.630	2.659	0.000	3.309	2.581
C_{14}	Tetradecanes	4.646	2.897	0.000	3.320	2.812
C_{15}	Pentadecanes	5.078	3.433	0.000	3.629	3.333
C_{16}	Hexadecanes	4.460	3.249	0.000	3.187	3.154
C_{17}	Heptadecanes	3.938	3.063	0.000	2.814	2.973
C_{18}	Octadecanes	4.043	3.330	0.000	2.889	3.233
C_{19}	Nonadecanes	3.887	3.355	0.000	2.778	3.257
C_{20}	Eicosanes	3.298	2.976	0.000	2.357	2.889
C_{21}	Heneicosanes	3.133	2.992	0.000	2.239	2.905
C_{22}	Docosanes	2.781	2.783	0.000	1.987	2.702
C_{23}	Tricosanes	2.592	2.705	0.000	1.852	2.626
C_{24}	Tetracosanes	2.395	2.601	0.000	1.711	2.525
C_{25}	Pentacosanes	2.216	2.509	0.000	1.584	2.436
C_{26}	Hexacosanes	1.841	2.169	0.000	1.316	2.106
C_{27}	Heptacosanes	1.713	2.102	0.000	1.224	2.041
C_{28}	Octacosanes	1.693	2.156	0.000	1.210	2.093
C_{29}	Nonacosanes	1.717	2.265	0.000	1.227	2.199
C_{30}	Triacontanes	1.567	2.139	0.000	1.120	2.076
C_{31}	Hentriacontanes	1.478	2.085	0.000	1.056	2.024
C_{32}	Dotriacontanes	1.163	1.695	0.000	0.831	1.645
C_{33}	Tritriacontanes	1.079	1.622	0.000	0.771	1.575
C_{34}	Tetratriacontanes	0.974	1.508	0.000	0.696	1.464
C_{35}	Pentatriacontanes	0.793	1.264	0.000	0.566	1.227
C_{36+}	Hexatriacontanes plus	9.382	30.742	0.000	6.703	29.841
	Totals :	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means less than 0.0005.

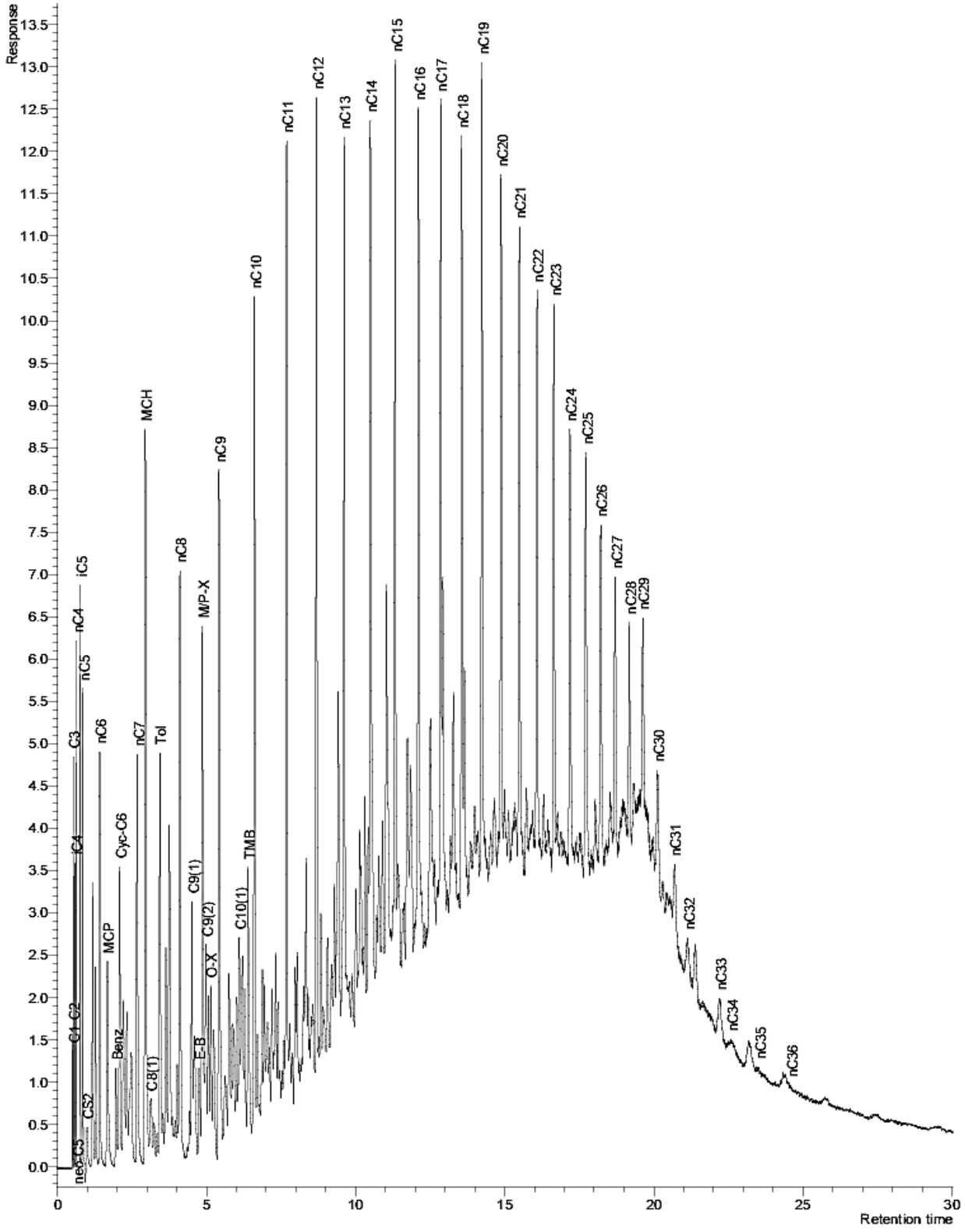
Compositional Analysis of Bottomhole Sample 13924-QA (1727.6 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.470	1.680	69.417
Molecular Weight (g mol-1)	313	98.7	312
Density at 60°F (g cm-3)	0.9035	0.7534	0.9030
C_{11+}			
Mole%	78.696	0.012	56.238
Molecular Weight (g mol-1)	358	149	358
Density at 60°F (g cm-3)	0.9178	0.7906	0.9177
C_{20+}			
Mole%	39.815		28.450
Molecular Weight (g mol-1)	508		508
Density at 60°F (g cm-3)	0.9573		0.9573
C_{36+}			
Mole %	9.382		6.703
Molecular Weight (g mol-1)	999		999
Density at 60°F (g cm-3)	1.0513		1.0513

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	305	23.0	224
Measured density at 60°F (g cm-3)	0.9002		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.797	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			155.9

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 13924-QA (1727.6 m TVDBRT)



Compositional Analysis of Bottomhole Sample 2587-EA (1727.6 m TVDBRT) to C36+

	Component	Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.000	0.000	0.415	0.118	0.023
N_2	Nitrogen	0.000	0.000	1.294	0.367	0.046
C_1	Methane	0.090	0.010	79.522	22.634	1.618
C_2	Ethane	0.090	0.010	6.544	1.922	0.257
C_3	Propane	0.240	0.040	4.513	1.454	0.286
iC_4	i-Butane	0.190	0.040	1.429	0.540	0.140
nC_4	n-Butane	0.380	0.070	1.752	0.767	0.199
C_5	neo-Pentane	0.000	0.000	0.023	0.007	0.002
iC_5	i-Pentane	0.510	0.120	1.132	0.690	0.222
nC_5	n-Pentane	0.490	0.120	0.775	0.570	0.183
C_6	Hexanes	1.360	0.390	0.977	1.251	0.480
	Me-Cyclo-pentane	0.530	0.150	0.184	0.433	0.162
	Benzene	0.210	0.050	0.072	0.168	0.059
	Cyclo-hexane	0.610	0.170	0.262	0.509	0.191
C_7	Heptanes	2.060	0.680	0.382	1.581	0.706
	Me-Cyclo-hexane	1.480	0.480	0.232	1.125	0.492
	Toluene	0.740	0.230	0.088	0.557	0.229
C_8	Octanes	3.180	1.190	0.206	2.335	1.188
	Ethyl-benzene	0.240	0.090	0.010	0.177	0.084
	Meta/Para-xylene	1.020	0.360	0.026	0.738	0.349
	Ortho-xylene	0.340	0.120	0.007	0.244	0.116
C_9	Nonanes	2.850	1.200	0.096	2.068	1.182
	Tri-Me-benzene	0.400	0.160	0.000	0.287	0.153
C_{10}	Decanes	3.930	1.840	0.043	2.826	1.791
C_{11}	Undecanes	4.030	1.950	0.016	2.889	1.892
C_{12}	Dodecanes	4.120	2.180	0.000	2.954	2.119
C_{13}	Tridecanes	4.620	2.660	0.000	3.308	2.579
C_{14}	Tetradecanes	4.590	2.870	0.000	3.286	2.782
C_{15}	Pentadecanes	5.050	3.420	0.000	3.618	3.321
C_{16}	Hexadecanes	4.420	3.220	0.000	3.164	3.129
C_{17}	Heptadecanes	3.960	3.090	0.000	2.840	2.998
C_{18}	Octadecanes	4.030	3.330	0.000	2.887	3.229
C_{19}	Nonadecanes	3.640	3.150	0.000	2.607	3.055
C_{20}	Eicosanes	3.280	2.970	0.000	2.352	2.881
C_{21}	Heneicosanes	3.110	2.970	0.000	2.226	2.886
C_{22}	Docosanes	2.840	2.840	0.000	2.031	2.760
C_{23}	Tricosanes	2.610	2.730	0.000	1.872	2.653
C_{24}	Tetracosanes	2.390	2.600	0.000	1.713	2.526
C_{25}	Pentacosanes	2.180	2.470	0.000	1.563	2.402
C_{26}	Hexacosanes	1.920	2.270	0.000	1.377	2.202
C_{27}	Heptacosanes	1.680	2.070	0.000	1.205	2.008
C_{28}	Octacosanes	1.720	2.190	0.000	1.229	2.124
C_{29}	Nonacosanes	1.710	2.260	0.000	1.226	2.196
C_{30}	Triacontanes	1.590	2.170	0.000	1.135	2.104
C_{31}	Hentriacontanes	1.410	1.990	0.000	1.008	1.930
C_{32}	Dotriacontanes	1.260	1.840	0.000	0.901	1.782
C_{33}	Tritriacontanes	1.100	1.660	0.000	0.789	1.610
C_{34}	Tetratriacontanes	0.950	1.480	0.000	0.683	1.436
C_{35}	Pentatriacontanes	0.800	1.270	0.000	0.570	1.233
C_{36+}	Hexatriacontanes plus	10.050	30.830	0.000	7.199	30.005
	Totals :	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means less than 0.0005.

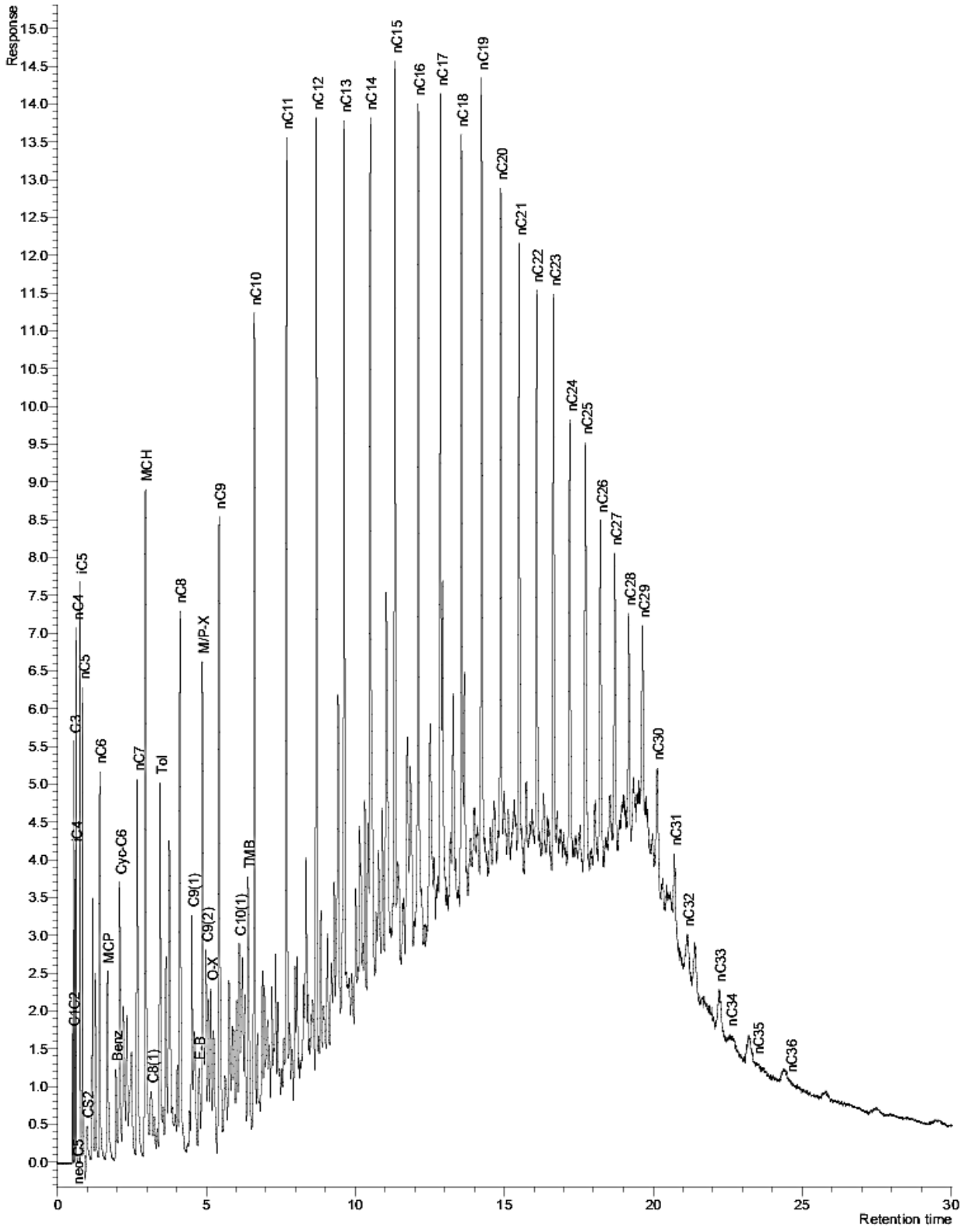
Compositional Analysis of Bottomhole Sample 2587-EA (1727.6 m TVDBRT) to C36+

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C_{7+}			
Mole%	96.650	1.624	69.680
Molecular Weight (g mol-1)	312	98.9	311
Density at 60°F (g cm-3)	0.9031	0.7536	0.9027
C_{11+}			
Mole%	79.060	0.016	56.632
Molecular Weight (g mol-1)	356	151	356
Density at 60°F (g cm-3)	0.9174	0.7921	0.9174
C_{20+}			
Mole%	40.600		29.079
Molecular Weight (g mol-1)	500		500
Density at 60°F (g cm-3)	0.9562		0.9562
C_{36+}			
Mole %	10.050		7.199
Molecular Weight (g mol-1)	936		936
Density at 60°F (g cm-3)	1.0484		1.0484

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
Average mole weight (g mol-1)	304	23.0	224
Measured density at 60°F (g cm-3)	0.9000		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.796	
Entrained water content (wt%)	0.02		
Flash Gas-oil ratio (scf/bbl)			154.8

Note: Compositional analysis data taken from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Oil ex Bottomhole Sample 2587-EA (1727.6 m TVDBRT)



Section D - PVT Analysis Data - Bottomhole Sample 2587-EA

Constant Composition Expansion at 167°F

Single-phase Fluid Properties

Saturation pressure (bubble-point pressure)	1057 psig
Average single phase compressibility (From 2877 psig to 1057 psig)	6.36 x 10 -6 psi-1
Density at saturation pressure	0.8364 g cm-3

Mean Single-phase Compressibility

Pressure Range Initial Pressure (psig)	Final Pressure (psig)	Mean Compressibility (psi-1) (1)
5000	3980	3.76 x 10 -6
3980	2877	4.42 x 10 -6
2877	1681	5.68 x 10 -6
1681	1057	7.67 x 10 -6

(1) Mean compressibility = (V2-V1) / [(V1+V2)/2] x 1/(P1 - P2)

Constant Composition Expansion at 167°F

Pressure (psig)		Relative Volume (1)	Density (g cm-3)	Instantaneous Compressibility (psi-1 x 10-6) (2)	Y-function (3)
5000		0.9799	0.8536	3.52	
4972		0.9800	0.8535	3.54	
3980		0.9837	0.8503	4.03	
2981		0.9880	0.8466	4.79	
2877	**Reservoir pressure**	**0.9885**	**0.8462**	**4.89**	
2485		0.9905	0.8445	5.33	
2227		0.9919	0.8433	5.69	
1980		0.9933	0.8421	6.11	
1883		0.9939	0.8416	6.29	
1779		0.9946	0.8410	6.51	
1681		0.9952	0.8405	6.73	
1581		0.9959	0.8399	6.98	
1476		0.9967	0.8393	7.27	
1432		0.9970	0.8390	7.41	
1377		0.9974	0.8386	7.58	
1329		0.9977	0.8383	7.74	
1278		0.9981	0.8380	7.93	
1224		0.9986	0.8376	8.14	
1175		0.9990	0.8373	8.34	
1130		0.9994	0.8370	8.53	
1081		0.9998	0.8366	8.76	
1067		0.9999	0.8365	8.83	
1057	**Saturation pressure**	**1.0000**	**0.8364**		
1001		1.0158			3.478
952		1.0317			3.444
902		1.0497			3.409
852		1.0699			3.376
802		1.0936			3.341
702		1.1514			3.274
602		1.2299			3.207
500		1.3449			3.141
400		1.5147			3.077
300		1.7983			3.015

(1) Relative Volume = V / Vsat ie. volume at indicated pressure per volume at saturation pressure.
(2) Instantaneous compressibility = (V2-V1) / V1 x 1/(P1-P2)
(3) Y-function = (Psat - P) / ((Pabs)(V/Vsat - 1)).

Partial Constant Composition Expansion at 140°F

Pressure (psig)		Relative Volume (1)	Density (g cm-3)	Instantaneous Compressibility (psi-1 x 10-6) (2)
5000		0.9816	0.8613	3.51
3975		0.9854	0.8581	3.87
2982		0.9894	0.8546	4.39
2750		0.9904	0.8537	4.55
2473		0.9917	0.8526	4.76
2227		0.9929	0.8516	4.98
1978		0.9941	0.8505	5.25
1875		0.9947	0.8500	5.37
1772		0.9952	0.8495	5.51
1685		0.9957	0.8491	5.63
1579		0.9963	0.8486	5.79
1477		0.9969	0.8481	5.97
1378		0.9975	0.8476	6.15
1275		0.9982	0.8470	6.37
1222		0.9985	0.8468	6.48
1175		0.9988	0.8465	6.60
1130		0.9991	0.8462	6.71
1078		0.9995	0.8459	6.85
1028		0.9998	0.8456	7.00
1009		1.0000	0.8455	7.06
1002	**Saturation pressure**	**1.0000**	**0.8455**	

(1) Relative Volume = V / Vsat ie. volume at indicated pressure per volume at saturation pressure.
(2) Instantaneous compressibility = (V2-V1) / V1 x 1/(P1-P2)

Partial Constant Composition Expansion at 104°F

Pressure (psig)		Relative Volume (1)	Density (g cm-3)	Instantaneous Compressibility (psi-1 x 10-6) (2)
5000		0.9843	0.8719	2.60
3977		0.9871	0.8694	2.97
2964		0.9903	0.8666	3.54
2721		0.9912	0.8658	3.72
2476		0.9921	0.8650	3.94
2228		0.9931	0.8641	4.19
1974		0.9942	0.8632	4.51
1718		0.9954	0.8621	4.90
1477		0.9966	0.8611	5.36
1383		0.9971	0.8606	5.57
1278		0.9977	0.8601	5.84
1184		0.9983	0.8596	6.11
1073		0.9990	0.8590	6.48
1015		0.9994	0.8587	6.70
969		0.9997	0.8584	6.89
935		0.9999	0.8582	7.04
926	**Saturation pressure**	**1.0000**	**0.8582**	

(1) Relative Volume = V / Vsat ie. volume at indicated pressure per volume at saturation pressure.
(2) Instantaneous compressibility = (V2-V1) / V1 x 1/(P1-P2)

Differential Vaporisation at 167°F

Pressure (psig)	Solution Gas-Oil Ratio Rs(1)	Relative Oil Volume Bod(2)	Relative Total Volume Btd(3)	Density (g cm-3)	Deviation Factor (Z)	Gas Formation Volume Factor (4)	Incremental Gas Gravity (Air = 1.000)
1057	**147**	**1.101**	**1.101**	**0.8364**		**Saturation Pressure**	
900	127	1.095	1.161	0.8391	0.938	0.01821	0.611
650	96	1.084	1.316	0.8434	0.951	0.02541	0.609
400	65	1.074	1.684	0.8478	0.966	0.04138	0.621
200	37	1.064	2.659	0.8516	0.981	0.08116	0.656
100	23	1.059	4.456	0.8539	0.990	0.15319	0.701
0	0	1.048		0.8580			1.010
	At 60°F =	1.000					

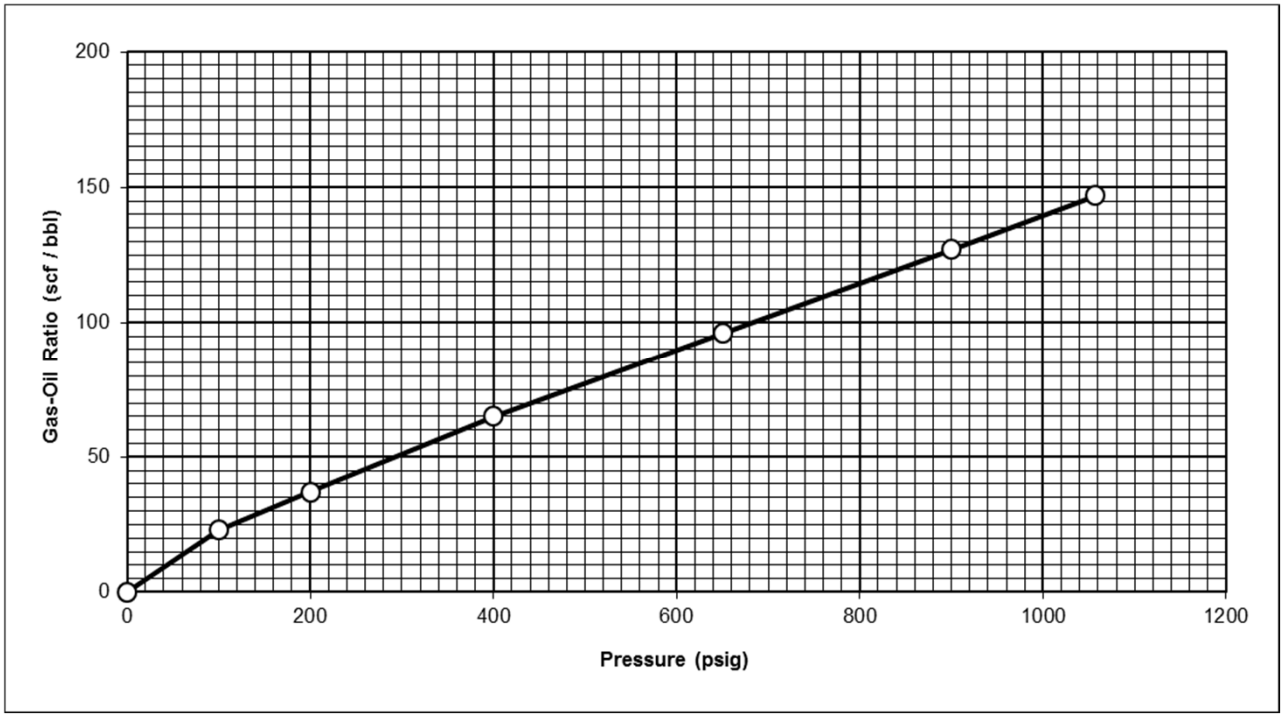
Residual Oil Properties

Density of residual oil	0.8990	g cm-3 at 60°F
API	25.7	

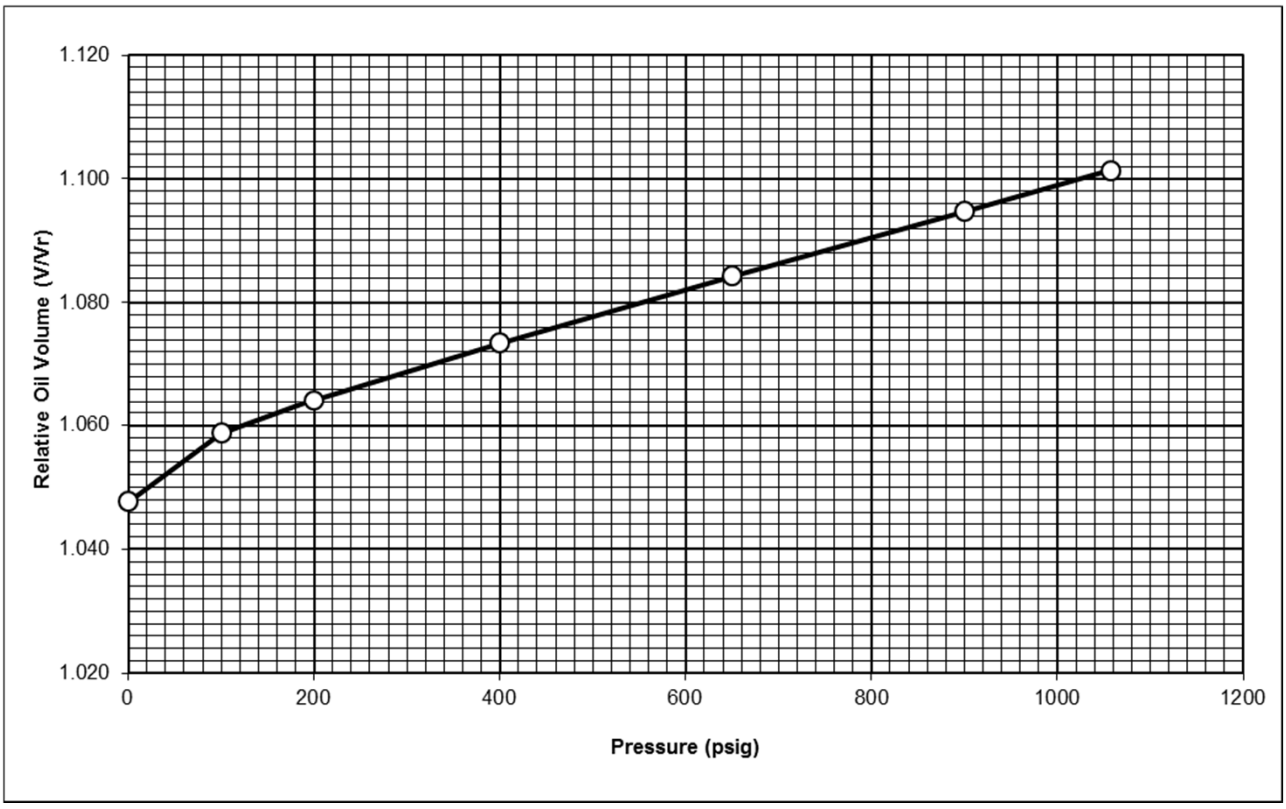
(1) GOR in cubic feet of gas at 14.73 psia and 60°F per barrel of residual oil at 60°F.
(2) Volume of oil at indicated pressure and temperature per volume of residual oil at 60°F.
(3) Volume of oil plus liberated gas at indicated pressure and temperature per volume of residual oil at 60°F.
(4) Volume of gas at indicated pressure and temperature per volume at 14.73 psia and 60°F.

Graphs from Differential Vaporisation at 167°F

Solution Gas-Oil Ratio v Pressure



Relative Oil Volume v Pressure



Compositional Analysis of Differential Vaporisation Gases to C11+

	Sample I.D.						
	Test Stage	1	2	3	4	5	6
	Stage Pressure (psig)	900	650	400	200	100	0
	Component (Mole%)						
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon Dioxide	0.276	0.313	0.379	0.483	0.597	0.741
N_2	Nitrogen	3.515	2.257	1.220	0.569	0.244	0.093
C_1	Methane	91.851	92.421	91.666	88.267	83.470	57.437
C_2	Ethane	2.518	2.954	3.922	5.955	8.548	16.743
C_3	Propane	0.925	1.063	1.509	2.525	3.810	11.369
iC_4	i-Butane	0.192	0.228	0.315	0.549	0.878	3.157
nC_4	n-Butane	0.206	0.248	0.328	0.560	0.900	3.553
C_5	Neo-Pentane	0.004	0.003	0.005	0.007	0.011	0.045
iC_5	i-Pentane	0.107	0.124	0.167	0.275	0.430	1.903
nC_5	n-Pentane	0.071	0.083	0.109	0.180	0.276	1.263
C_6	Hexanes	0.101	0.107	0.134	0.221	0.326	1.525
	M-C-Pentane	0.019	0.019	0.024	0.041	0.058	0.273
	Benzene	0.007	0.008	0.009	0.016	0.022	0.104
	Cyclohexane	0.028	0.029	0.035	0.059	0.082	0.376
C_7	Heptanes	0.050	0.047	0.056	0.092	0.124	0.551
	M-C-Hexane	0.028	0.027	0.034	0.055	0.074	0.315
	Toluene	0.013	0.012	0.013	0.023	0.031	0.118
C_8	Octanes	0.033	0.027	0.036	0.057	0.068	0.264
	E-Benzene	0.002	0.001	0.002	0.003	0.003	0.010
	M/P-Xylene	0.006	0.004	0.005	0.008	0.008	0.028
	O-Xylene	0.002	0.001	0.002	0.002	0.002	0.007
C_9	Nonanes	0.021	0.015	0.019	0.030	0.030	0.099
	1,2,4-TMB	0.000	0.000	0.000	0.000	0.000	0.000
C_{10}	Decanes	0.017	0.007	0.009	0.017	0.008	0.026
C_{11+}	Undecanes plus	0.008	0.002	0.002	0.006	0.000	0.001
	Totals :	100.000	100.000	100.000	100.000	100.000	100.000
	Calculated Gas Properties						
	Gas Gravity (Air = 1.000)	0.611	0.609	0.621	0.656	0.701	1.010

Note: 0.000 means < 0.0005.

Compositional Analysis of Differential Vaporisation Residual Oil to C36+

	Component	Mole %	Weight %
H_2	Hydrogen	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000
CO_2	Carbon Dioxide	0.000	0.000
N_2	Nitrogen	0.000	0.000
C_1	Methane	0.043	0.002
C_2	Ethane	0.214	0.021
C_3	Propane	0.589	0.086
iC_4	i-Butane	0.377	0.073
nC_4	n-Butane	0.662	0.128
C_5	Neo-Pentane	0.000	0.000
iC_5	i-Pentane	0.719	0.172
nC_5	n-Pentane	0.643	0.154
C_6	Hexanes	1.609	0.461
	M-C-Pentane	0.561	0.157
	Benzene	0.211	0.055
	Cyclohexane	0.654	0.183
C_7	Heptanes	2.041	0.679
	M-C-Hexane	1.563	0.510
	Toluene	0.664	0.203
C_8	Octanes	3.080	1.169
	E-Benzene	0.237	0.084
	M/P-Xylene	0.888	0.313
	O-Xylene	0.261	0.092
C_9	Nonanes	2.919	1.244
	1,2,4-TMB	0.340	0.136
C_{10}	Decanes	3.895	1.841
C_{11}	Undecanes	3.903	1.906
C_{12}	Dodecanes	4.057	2.170
C_{13}	Tridecanes	4.542	2.641
C_{14}	Tetradecanes	4.476	2.826
C_{15}	Pentadecanes	5.047	3.455
C_{16}	Hexadecanes	4.367	3.221
C_{17}	Heptadecanes	3.874	3.051
C_{18}	Octadecanes	3.984	3.322
C_{19}	Nonadecanes	3.713	3.245
C_{20}	Eicosanes	3.356	3.067
C_{21}	Heneicosanes	3.041	2.940
C_{22}	Docosanes	2.790	2.827
C_{23}	Tricosanes	2.544	2.688
C_{24}	Tetracosanes	2.241	2.465
C_{25}	Pentacosanes	2.112	2.421
C_{26}	Hexacosanes	1.882	2.245
C_{27}	Heptacosanes	1.682	2.091
C_{28}	Octacosanes	1.665	2.147
C_{29}	Nonacosanes	1.620	2.164
C_{30}	Triacontanes	1.530	2.114
C_{31}	Hentriacontanes	1.443	2.062
C_{32}	Dotriacontanes	1.190	1.756
C_{33}	Tritriacontanes	1.049	1.596
C_{34}	Tetratriacontanes	0.926	1.452
C_{35}	Pentatriacontanes	0.799	1.291
C_{36+}	Hexatriacontanes plus	9.997	31.074
	Totals :	100.000	100.000

Note: 0.000 means < 0.0005.

Compositional Analysis of Differential Vaporisation Residual Oil to C36+

Calculated Residue Properties	
C_{7+}	
Mole%	95.144
Molecular Weight (g mol-1)	313
Density at 60°F (g cm-3)	0.9033
C_{11+}	
Mole%	77.830
Molecular Weight (g mol-1)	357
Density at 60°F (g cm-3)	0.9178
C_{20+}	
Mole%	39.867
Molecular Weight (g mol-1)	501
Density at 60°F (g cm-3)	0.9568
C_{36+}	
Mole %	9.997
Molecular Weight (g mol-1)	936
Density at 60°F (g cm-3)	1.0484

Calculated Whole Sample Properties	
Average mole weight (g mol-1)	301
Density at 60°F (g cm-3) [Measured]	0.8990
API	25.7

Differential Vaporisation Data Converted to Separator Test Conditions

Pressure (psig)		Oil Density (g cm-3)	Solution Gas/Oil (scf / bbl) Rs(1)	Formation Volume Factor Bo(1)	Gas Formation Volume Factor Bg(2)
5000		0.8535		1.081	
3980		0.8503		1.085	
2877	**Reservoir pressure**	**0.8461**		**1.091**	
2485		0.8445		1.093	
2227		0.8433		1.094	
1980		0.8420		1.096	
1883		0.8415		1.097	
1681		0.8404		1.098	
1581		0.8398		1.099	
1432		0.8389		1.100	
1377		0.8386		1.100	
1278		0.8380		1.101	
1224		0.8376		1.102	
1057	**Saturation pressure**	**0.8364**	**152**	**1.103**	
900		0.8391	131	1.097	0.01821
650		0.8434	100	1.086	0.02541
400		0.8478	69	1.075	0.04138
200		0.8516	41	1.066	0.08116
100		0.8539	27	1.061	0.15319

Notes:

(1) Differential data corrected to surface separator conditions of :-

Stage 1	80 psig and 149°F
Stage 2	7 psig and 194°F
Stage 3	0 psig and 60°F

Rs = Rsfb - (Rsdb - Rsd) x (Bofb / Bodb)

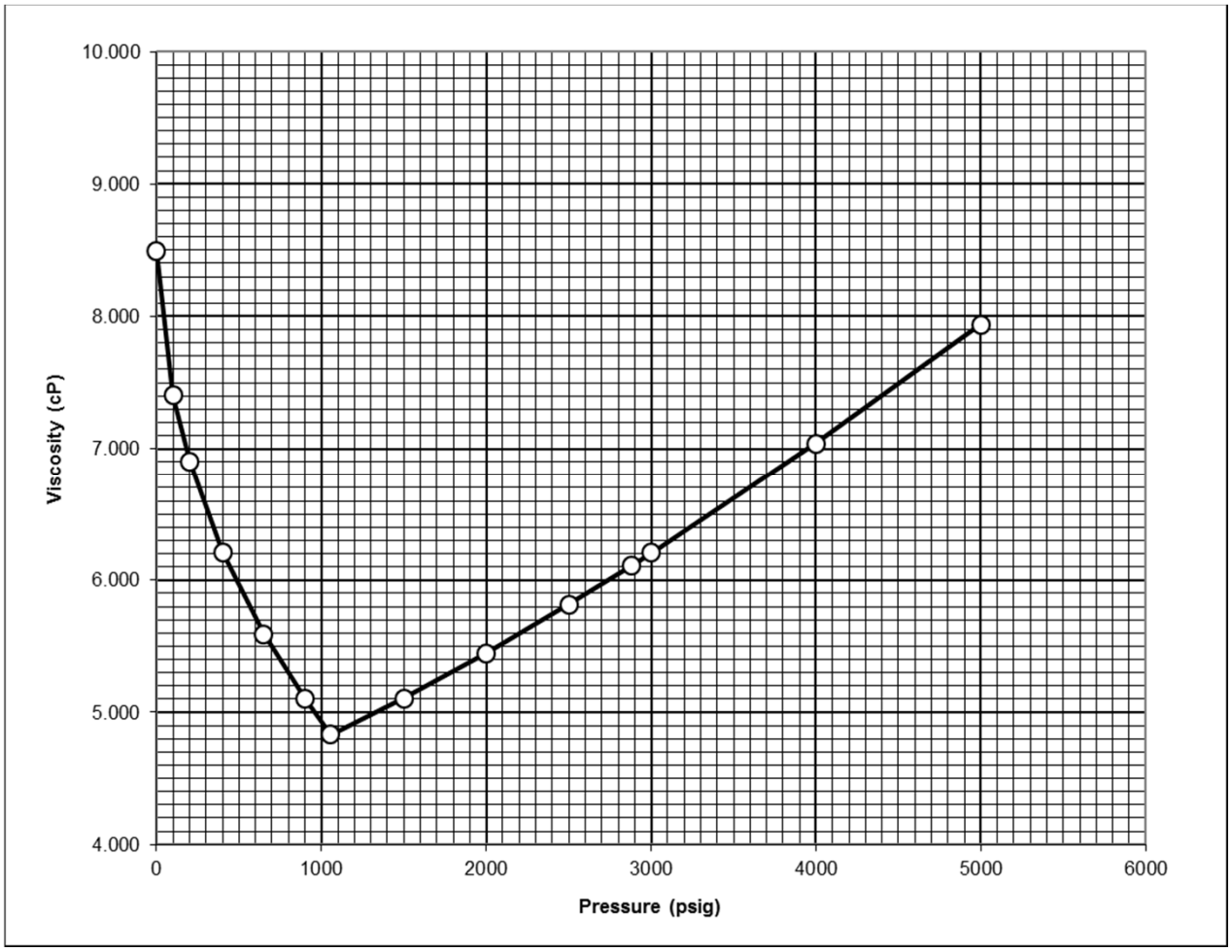
Bo = Bod x (Bofb/Bodb)

(2) Volume of gas at indicated pressure and temperature per volume at 14.73 psia and 60°F.

Reservoir Fluid Viscosity Data at 167°F

Pressure (psig)		Oil Viscosity (cP)	Calculated Gas Viscosity (cP) (1)	Oil / Gas Viscosity Ratio
5000		7.938		
4000		7.040		
3000		6.205		
2877	**Reservoir pressure**	**6.108**		
2500		5.816		
2000		5.448		
1500		5.106		
1057	**Saturation pressure**	**4.829**		
900		5.099	0.0140	363.6
650		5.592	0.0136	410.5
400		6.211	0.0132	469.8
200		6.899	0.0128	538.3
100		7.405	0.0125	592.5
0		8.490		

Reservoir Fluid Viscosity v Pressure at 167°F

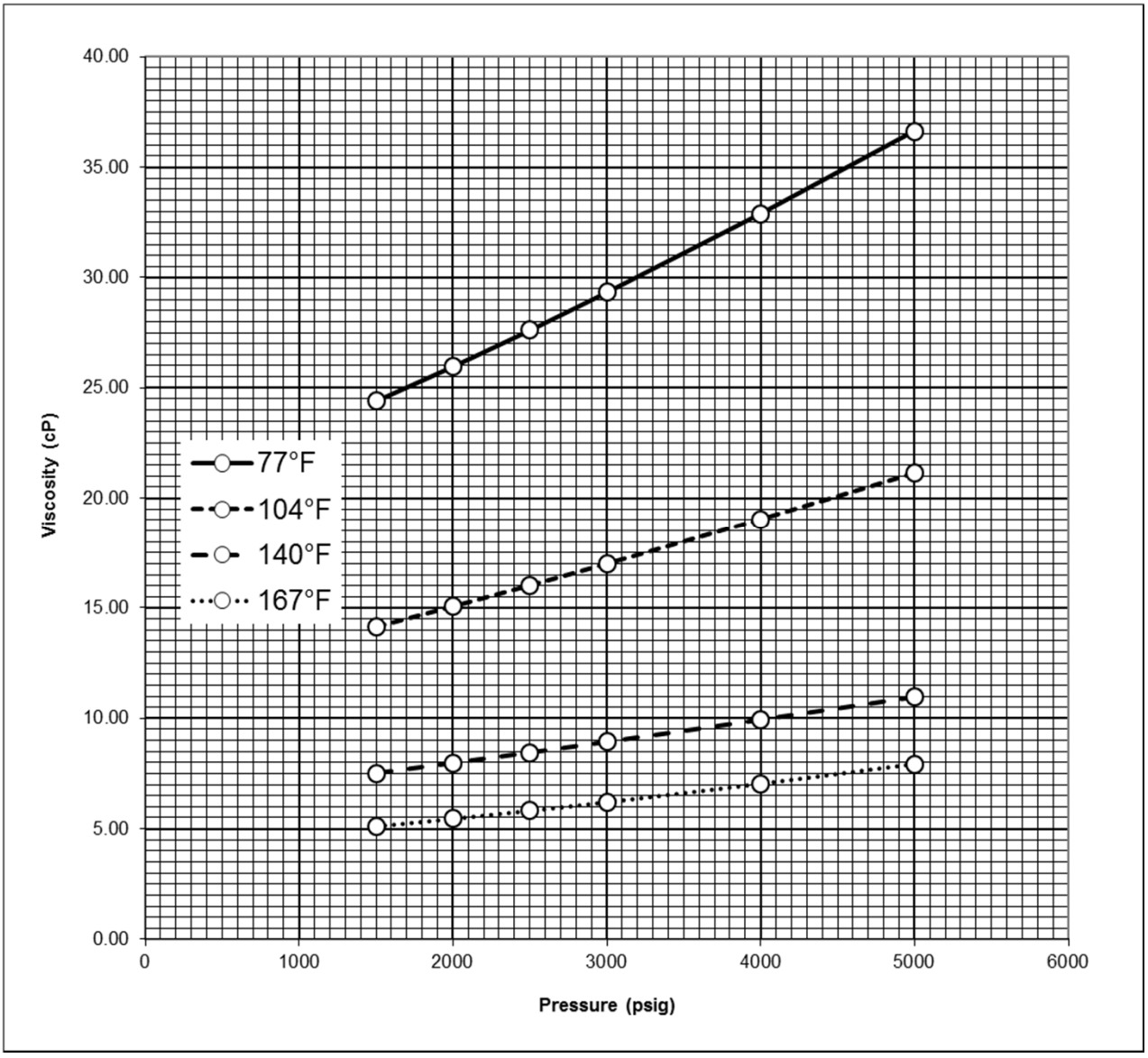


(1) Calculated using the method of Lee, Gonzales and Eakin, JPT, Aug 1966.

Single-Phase Reservoir Fluid Viscosity Data at Various Temperatures

Pressure (psig)	Oil Viscosity at 77°F (cP)	Oil Viscosity at 104°F (cP)	Oil Viscosity at 140°F (cP)	Oil Viscosity at 167°F (cP)
5000	36.64	21.15	10.97	7.938
4000	32.90	19.04	9.938	7.040
3000	29.33	17.00	8.940	6.205
2500	27.62	16.02	8.455	5.816
2000	25.97	15.06	7.982	5.448
1500	24.39	14.14	7.522	5.106

Single-Phase Reservoir Fluid Viscosity v Pressure



Atmospheric Flash Data

Pressure (psig)	Temperature (°F)	Gas-Oil Ratio Rs (1)	Flashed Oil Density (g cm-3)	Formation Volume Factor Bob (2)	Formation Volume Factor Bo (3)	Flashed Gas Gravity (Air = 1.000)
1057	**167**		**Saturation Pressure**			
0	60	140	0.8974 (API = 26.0°)	1.095	1.083	0.678

(1) GOR in cubic feet of gas at 14.73 psia and 60°F per barrel of stocktank oil at 60°F.
(2) Volume of saturated oil at 1057 psig and 167°F per volume of stocktank oil at 60°F.
(3) Volume of undersaturated oil at 2877 psig and 167°F per volume of stocktank oil at 60°F.

Compositional Analysis of Atmospheric Flash Gas to C11+

	Component (Mole%)	
H_2	Hydrogen	0.000
H_2S	Hydrogen Sulphide	0.000
CO_2	Carbon Dioxide	0.374
N_2	Nitrogen	1.404
C_1	Methane	85.313
C_2	Ethane	6.500
C_3	Propane	3.551
iC_4	i-Butane	0.889
nC_4	n-Butane	0.938
C_5	Neo-Pentane	0.012
iC_5	i-Pentane	0.414
nC_5	n-Pentane	0.246
C_6	Hexanes	0.209
	M-C-Pentane	0.031
	Benzene	0.011
	Cyclohexane	0.034
C_7	Heptanes	0.038
	M-C-Hexane	0.017
	Toluene	0.005
C_8	Octanes	0.008
	E-Benzene	0.000
	M/P-Xylene	0.001
	O-Xylene	0.000
C_9	Nonanes	0.003
	1,2,4-TMB	0.000
C_{10}	Decanes	0.002
C_{11+}	Undecanes plus	0.000
	Totals :	100.000

	Calculated Gas Properties	
	Gas Gravity (Air = 1.000)	0.678
	Mole weight (g mol-1)	19.6

Note: 0.000 means < 0.0005.

Compositional Analysis of Atmospheric Flash Stocktank Oil to C36+

	Component	Mole %	Weight %
H_2	Hydrogen	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000
CO_2	Carbon Dioxide	0.000	0.000
N_2	Nitrogen	0.000	0.000
C_1	Methane	0.148	0.008
C_2	Ethane	0.205	0.020
C_3	Propane	0.528	0.077
iC_4	i-Butane	0.362	0.070
nC_4	n-Butane	0.641	0.124
C_5	Neo-Pentane	0.000	0.000
iC_5	i-Pentane	0.741	0.177
nC_5	n-Pentane	0.650	0.155
C_6	Hexanes	1.612	0.460
	M-C-Pentane	0.565	0.158
	Benzene	0.226	0.058
	Cyclohexane	0.670	0.187
C_7	Heptanes	2.064	0.686
	M-C-Hexane	1.591	0.518
	Toluene	0.724	0.221
C_8	Octanes	3.103	1.175
	E-Benzene	0.242	0.085
	M/P-Xylene	0.926	0.326
	O-Xylene	0.278	0.098
C_9	Nonanes	2.862	1.217
	1,2,4-TMB	0.430	0.171
C_{10}	Decanes	3.835	1.809
C_{11}	Undecanes	3.882	1.892
C_{12}	Dodecanes	4.028	2.150
C_{13}	Tridecanes	4.489	2.604
C_{14}	Tetradecanes	4.446	2.800
C_{15}	Pentadecanes	5.024	3.431
C_{16}	Hexadecanes	4.279	3.149
C_{17}	Heptadecanes	3.885	3.053
C_{18}	Octadecanes	3.961	3.295
C_{19}	Nonadecanes	3.712	3.236
C_{20}	Eicosanes	3.303	3.011
C_{21}	Heneicosanes	3.088	2.979
C_{22}	Docosanes	2.746	2.777
C_{23}	Tricosanes	2.512	2.648
C_{24}	Tetracosanes	2.317	2.543
C_{25}	Pentacosanes	2.057	2.352
C_{26}	Hexacosanes	1.847	2.198
C_{27}	Heptacosanes	1.740	2.157
C_{28}	Octacosanes	1.630	2.097
C_{29}	Nonacosanes	1.602	2.136
C_{30}	Triacontanes	1.531	2.111
C_{31}	Hentriacontanes	1.423	2.028
C_{32}	Dotriacontanes	1.177	1.733
C_{33}	Tritriacontanes	1.051	1.596
C_{34}	Tetratriacontanes	0.938	1.468
C_{35}	Pentatriacontanes	0.769	1.240
C_{36+}	Hexatriacontanes plus	10.160	31.516
	Totals :	100.000	100.000

Note: 0.000 means < 0.0005.

Compositional Analysis of Atmospheric Flash Stocktank Oil to C36+

Calculated Residue Properties	
C_{7+}	
Mole%	95.113
Molecular Weight (g mol-1)	314
Density at 60°F (g cm-3)	0.9042
C_{11+}	
Mole%	77.597
Molecular Weight (g mol-1)	358
Density at 60°F (g cm-3)	0.9187
C_{20+}	
Mole%	39.891
Molecular Weight (g mol-1)	504
Density at 60°F (g cm-3)	0.9576
C_{36+}	
Mole %	10.160
Molecular Weight (g mol-1)	936
Density at 60°F (g cm-3)	1.0484

Calculated Whole Sample Properties	
Average mole weight (g mol-1)	302
Density at 60°F (g cm-3) [Measured]	0.8974
API	26.0

Separator Test Data

Pressure (psig)	Temperature (°F)	Gas-Oil Ratio (1)	Gas-Oil Ratio Rsfb (2)	Oil Density (g cm-3)	Formation Volume Factor Bofb (3)	Separation Volume Factor (4)	Gas Gravity of flashed gas (Air = 1.000)
1057	**167**	**-**		**0.8364**	**1.103**	**Saturation Pressure**	
80	149	127	133	0.8635		1.047	0.679
7	194	18	19	0.8580		1.048	1.121
0	60	0	0	0.8994		1.000	*
			____	(API = 25.7°)			
		Total Rsfb:	152				

* Insufficient volume of gas evolved to accurately determine gas gravity or hydrocarbon composition.

(1) GOR in cubic feet of gas at 14.73 psia and 60°F per barrel of oil at indicated pressure and temperature.
(2) GOR in cubic feet of gas at 14.73 psia and 60°F per barrel of stocktank oil at 60°F.
(3) Volume of saturated oil at 1057 psig and 167°F per volume of stocktank oil at 60°F.
(4) Volume of oil at indicated pressure and temperature per volume of stocktank oil at 60°F.

Compositional Analysis of Separator Test Gases to C11+

	Sample I.D.		
	Test Stage	1	2
	Stage Pressure (psig)	80	7
	Component (Mole%)		
H_2	Hydrogen	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000
CO_2	Carbon Dioxide	0.336	0.625
N_2	Nitrogen	1.549	0.480
C_1	Methane	86.268	54.013
C_2	Ethane	5.684	12.178
C_3	Propane	3.081	12.218
iC_4	i-Butane	0.740	4.507
nC_4	n-Butane	0.807	5.491
C_5	Neo-Pentane	0.010	0.075
iC_5	i-Pentane	0.411	3.222
nC_5	n-Pentane	0.268	2.087
C_6	Hexanes	0.312	2.357
	M-C-Pentane	0.056	0.392
	Benzene	0.022	0.142
	Cyclohexane	0.080	0.520
C_7	Heptanes	0.123	0.737
	M-C-Hexane	0.072	0.384
	Toluene	0.032	0.182
C_8	Octanes	0.069	0.278
	E-Benzene	0.003	0.007
	M/P-Xylene	0.010	0.018
	O-Xylene	0.003	0.004
C_9	Nonanes	0.035	0.068
	1,2,4-TMB	0.000	0.000
C_{10}	Decanes	0.018	0.010
C_{11+}	Undecanes plus	0.011	0.005
	Totals :	100.000	100.000
	Calculated Gas Properties		
	Gas Gravity (Air = 1.000)	0.680	1.120

Note: 0.000 means < 0.0005.

Compositional Analysis of Separator Test Stocktank Oil to C36+

	Component	Mole %	Weight %
H_2	Hydrogen	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000
CO_2	Carbon Dioxide	0.000	0.000
N_2	Nitrogen	0.000	0.000
C_1	Methane	0.040	0.000
C_2	Ethane	0.080	0.010
C_3	Propane	0.260	0.040
iC_4	i-Butane	0.210	0.040
nC_4	n-Butane	0.440	0.080
C_5	Neo-Pentane	0.000	0.000
iC_5	i-Pentane	0.580	0.140
nC_5	n-Pentane	0.550	0.130
C_6	Hexanes	1.470	0.410
	M-C-Pentane	0.540	0.150
	Benzene	0.200	0.050
	Cyclohexane	0.640	0.180
C_7	Heptanes	2.030	0.670
	M-C-Hexane	1.550	0.500
	Toluene	0.660	0.200
C_8	Octanes	3.170	1.190
	E-Benzene	0.240	0.080
	M/P-Xylene	0.850	0.300
	O-Xylene	0.250	0.090
C_9	Nonanes	3.030	1.270
	1,2,4-TMB	0.430	0.170
C_{10}	Decanes	3.920	1.830
C_{11}	Undecanes	3.960	1.910
C_{12}	Dodecanes	4.090	2.160
C_{13}	Tridecanes	4.650	2.670
C_{14}	Tetradecanes	4.460	2.780
C_{15}	Pentadecanes	5.060	3.410
C_{16}	Hexadecanes	4.370	3.180
C_{17}	Heptadecanes	3.930	3.050
C_{18}	Octadecanes	4.010	3.300
C_{19}	Nonadecanes	3.790	3.260
C_{20}	Eicosanes	3.370	3.040
C_{21}	Heneicosanes	3.100	2.960
C_{22}	Docosanes	2.780	2.780
C_{23}	Tricosanes	2.580	2.690
C_{24}	Tetracosanes	2.330	2.520
C_{25}	Pentacosanes	2.110	2.390
C_{26}	Hexacosanes	1.920	2.260
C_{27}	Heptacosanes	1.730	2.120
C_{28}	Octacosanes	1.660	2.110
C_{29}	Nonacosanes	1.650	2.180
C_{30}	Triacontanes	1.610	2.190
C_{31}	Hentriacontanes	1.380	1.950
C_{32}	Dotriacontanes	1.190	1.730
C_{33}	Tritriacontanes	1.060	1.590
C_{34}	Tetratriacontanes	0.970	1.500
C_{35}	Pentatriacontanes	0.850	1.350
C_{36+}	Hexatriacontanes plus	10.250	31.390
	Totals :	100.000	100.000

Note: 0.000 means < 0.0005.

Compositional Analysis of Separator Test Stocktank Oil to C36+

Calculated Residue Properties	
C_{7+}	
Mole%	96.370
Molecular Weight (g mol-1)	314
Density at 60°F (g cm-3)	0.9039
C_{11+}	
Mole%	78.860
Molecular Weight (g mol-1)	358
Density at 60°F (g cm-3)	0.9184
C_{20+}	
Mole%	40.540
Molecular Weight (g mol-1)	503
Density at 60°F (g cm-3)	0.9573
C_{36+}	
Mole %	10.250
Molecular Weight (g mol-1)	936
Density at 60°F (g cm-3)	1.0484

Calculated Whole Sample Properties	
Average mole weight (g mol-1)	305
Density at 60°F (g cm-3) [Measured]	0.8994
API	25.7

Section E - Summary of Separator Samples Received and Validation Data

Reported Well and Sampling Information - Clean Up Flow and Main Flow 1

Reservoir and Well Information

Field	Clair
Well	206/12-1E
Reservoir fluid	Oil
Reservoir pressure	
Reservoir temperature	
Installation	Paul B Loyd Jnr
Test	DST #1
Perforations	1937.4 - 1964.52 m MDRT
Formation	Jurassic

Sampling Information

Date sampled	06/02/2015
Time sampled	15:20
Type of samples	Separator
Mud system used	OBM
Sampling company	Schlumberger
Bottomhole pressure	
Bottomhole temperature	
Wellhead pressure	304 psia
Wellhead temperature	19°C
Separator pressure	178.7 psia
Separator temperature	26.8°C
Choke size	26/64"
Stable since	
H2S	0 ppm
CO2	0.0%
Gas rate (std cond)	05. MMscf/d
Gas gravity (air =1)	0.646
Fb factor	
Fpv factor	
Oil rate (std cond)	
Oil rate (sep cond)	977.9 bbl/d
Oil gravity 60/60	0.868
Water rate	05. bbl/d
BSW	0%

Comments: Well and sampling information for separator set A00144 and 5384-EA as reported by Schlumberger and BP Exploration Operating Co Ltd.

Reported Well and Sampling Information - Main Flow 2

Reservoir and Well Information

Field	Clair
Well	206/12-1E
Reservoir fluid	Oil
Reservoir pressure	
Reservoir temperature	
Installation	Paul B Loyd Jnr
Test	DST #1
Sampling depth	1937.4 - 1964.52 m MDRT
Formation	Jurassic

Sampling Information

Date sampled	09/02/2015
Time sampled	20:35
Type of samples	Separator
Mud system used	OBM
Sampling company	Schlumberger
Bottomhole pressure	
Bottomhole temperature	
Wellhead pressure	329 psia
Wellhead temperature	15°C
Separator pressure	187 psia
Separator temperature	27°C
Choke size	24/64"
Stable since	
H2S	0 ppm
CO2	1.0%
Gas rate (std cond)	0.04 MMscf/d
Gas gravity (air =1)	0.650
Fb factor	
Fpv factor	
Oil rate (std cond)	
Oil rate (sep cond)	1047.7 bbl/d
Oil gravity 60/60	0.866
Water rate	
BSW	0%

Comments: Well and sampling information for separator set A00127 and 5697-EA as reported by Schlumberger and BP Exploration Operating Co Ltd.

Summary of Clean Up Flow and Main Flow 1 Separator Samples Received and Validation Data

Separator Gases - Clean Up Flow and Main Flow 1

Sample Number	Cylinder Number	Sample:- Date	Time	Sampling:- Pressure (psig)	Temperature (°F)	Laboratory opening:- Pressure (psig)	Temperature (°F)	Free Liquid Hydro-C. (cm3)	Air Content (Mole%)
1.23	A01034	05/02/15	22:50	171	94	194	119	0	0.146
1.30	A01439	05/02/15	23:30	171	94	197	119	0	0.182
1.37	A01055	06/02/15	00:10	172	94	194	119	0	0.264
1.71	1768	06/02/15	12:09	166	80	183	101	0	0.140
1.73	A01453	06/02/15	12:45	167	81	183	101	0	0.177
1.79	A01416	06/02/15	13:20	164	81	184	101	0	0.411
1.81	A01422	06/02/15	14:00	164	81	176	101	0	0.124
1.87	A01443	06/02/15	14:47	165	81	180	101	0	0.183
1.93	A01429	06/02/15	15:20	164	80	184	101	0	0.214

Separator Liquids - Clean Up Flow and Main Flow 1

Sample Number	Cylinder Number	Sample:- Date	Time	Separator:- Pressure (psig)	Temperature (°F)	Laboratory bubble point:- Pressure (psig)	Temperature (°F)	Free Water Recovered (cm3)	Sample Volume (cm3)
1.22	4504-EA	05/02/15	22:50	171	94	146	66	3	583
1.29	4516-EA	05/02/15	23:30	171	94	149	66	1	578
1.36	5205-EA	06/02/15	00:10	172	94	146	66	0	628
1.70	4337-EA	06/02/15	12:09	166	80	145	68	0	580
1.72	5727-EA	06/02/15	12:45	167	81	137	62	0	593
1.78	4603-EA	06/02/15	13:20	164	81	139	60	0	585
1.80	5684-EA	06/02/15	14:00	164	81	143	68	0	630
1.86	4479-EA	06/02/15	14:47	165	81	150	68	0	593
1.92	14110-QA	06/02/15	15:20	164	80	141	70	0	588

Summary of Main Flow 2 Separator Samples Received and Validation Data

Separator Gases - Main Flow 2

Sample Number	Cylinder Number	Sample:- Date	Time	Separator:- Pressure (psig)	Temperature (°F)	Laboratory opening:- Pressure (psig)	Temperature (°F)	Free Liquid Hydro-C. (cm3)	Air Content (Mole%)
2.49	A01029	09/02/15	20:35	172	81	179	101	0	0.208
2.55	A01529	09/02/15	21:00	166	81	185	101	0	0.161
2.57	A01442	09/02/15	21:18	171	81	188	101	0	0.170

Separator Liquids - Main Flow 2

Sample Number	Cylinder Number	Sample:- Date	Time	Separator:- Pressure (psig)	Temperature (°F)	Laboratory bubble point:- Pressure (psig)	Temperature (°F)	Free Water Recovered (cm3)	Sample Volume (cm3)
2.48	4654-EA	09/02/15	20:35	172	81	148	68	0	595
2.54	5216-EA	09/02/15	21:00	166	81	141	69	0	591
2.56	5598-EA	09/02/15	21:18	171	81	147	70	0	590

Compositional Analysis of Clean Up and Main Flow 1 Separator Gases to C11+

	Sample Number	1.23	1.30	1.37	1.71
	Cylinder Number	A01034	A01439	A01055	1768
	Sample Date	05/02/15	05/02/15	06/02/15	06/02/15
	Sample Time	22:50	23:30	00:10	12:09
	Component	Mole%	Mole%	Mole%	Mole%
H_2	Hydrogen	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.415	0.414	0.395	0.394
N_2	Nitrogen	1.620	1.630	1.619	1.629
C_1	Methane	89.648	89.568	89.504	90.019
C_2	Ethane	4.770	4.785	4.778	4.705
C_3	Propane	1.956	1.994	2.114	1.843
iC_4	i-Butane	0.454	0.453	0.451	0.417
nC_4	n-Butane	0.466	0.466	0.464	0.420
C_5	neo-Pentane	0.006	0.006	0.006	0.005
iC_5	i-Pentane	0.223	0.220	0.221	0.194
nC_5	n-Pentane	0.140	0.138	0.138	0.120
C_6	Hexanes	0.144	0.147	0.143	0.121
	Me-Cyclo-pentane	0.024	0.025	0.024	0.020
	Benzene	0.008	0.009	0.009	0.007
	Cyclo-hexane	0.031	0.033	0.032	0.026
C_7	Heptanes	0.043	0.049	0.046	0.036
	Me-Cyclo-hexane	0.023	0.025	0.023	0.019
	Toluene	0.007	0.009	0.008	0.006
C_8	Octanes	0.016	0.019	0.017	0.013
	Ethyl-benzene	0.000	0.001	0.001	0.000
	Meta/Para-xylene	0.001	0.002	0.001	0.001
	Ortho-xylene	0.000	0.000	0.000	0.000
C_9	Nonanes	0.005	0.005	0.004	0.005
	1,2,4-TMB	0.000	0.000	0.000	0.000
C_{10}	Decanes	0.000	0.002	0.002	0.000
C_{11+}	Undecanes plus	0.000	0.000	0.000	0.000
	Totals :	100.000	100.000	100.000	100.000
	Sample Properties				
	Real Relative Density (Air=1.000 at 14.73 psia and 60°F)	0.637	0.638	0.638	0.632
	Molecular Weight (g mol-1)	18.4	18.4	18.4	18.2

Note: 0.000 means < 0.0005

Compositional Analysis of Clean Up and Main Flow 1 Separator Gases to C11+

	Sample Number	1.73	1.79	1.81	1.87	1.93
	Cylinder Number	A01453	A01416	A01422	A01443	A01429
	Sample Date	06/02/15	06/02/15	06/02/15	06/02/15	06/02/15
	Sample Time	12:45	13:20	14:00	14:47	15:20
	Component	Mole%	Mole%	Mole%	Mole%	Mole%
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.398	0.396	0.397	0.395	0.398
N_2	Nitrogen	1.631	1.620	1.621	1.634	1.626
C_1	Methane	89.989	89.904	89.924	89.820	89.956
C_2	Ethane	4.689	4.710	4.716	4.735	4.725
C_3	Propane	1.851	1.876	1.877	1.901	1.857
iC_4	i-Butane	0.418	0.423	0.426	0.422	0.421
nC_4	n-Butane	0.422	0.430	0.431	0.440	0.423
C_5	neo-Pentane	0.005	0.006	0.005	0.006	0.006
iC_5	i-Pentane	0.197	0.203	0.200	0.204	0.197
nC_5	n-Pentane	0.123	0.127	0.125	0.128	0.122
C_6	Hexanes	0.128	0.134	0.128	0.137	0.125
	Me-Cyclo-pentane	0.021	0.023	0.022	0.023	0.021
	Benzene	0.008	0.009	0.008	0.009	0.007
	Cyclo-hexane	0.028	0.031	0.028	0.032	0.027
C_7	Heptanes	0.040	0.044	0.039	0.045	0.038
	Me-Cyclo-hexane	0.022	0.025	0.022	0.025	0.020
	Toluene	0.007	0.009	0.007	0.009	0.007
C_8	Octanes	0.016	0.019	0.015	0.020	0.014
	Ethyl-benzene	0.000	0.001	0.001	0.001	0.001
	Meta/Para-xylene	0.001	0.002	0.001	0.002	0.001
	Ortho-xylene	0.000	0.001	0.000	0.001	0.000
C_9	Nonanes	0.005	0.006	0.006	0.008	0.005
	1,2,4-TMB	0.000	0.000	0.000	0.000	0.000
C_{10}	Decanes	0.001	0.001	0.001	0.003	0.002
C_{11+}	Undecanes plus	0.000	0.000	0.000	0.000	0.001
	Totals :	100.000	100.000	100.000	100.000	100.000
	Sample Properties					
	Real Relative Density (Air=1.000 at 14.73 psia and 60°F)	0.632	0.634	0.633	0.635	0.633
	Molecular Weight (g mol-1)	18.3	18.3	18.3	18.3	18.3

Note: 0.000 means < 0.0005

Compositional Analysis of Main Flow 2 Separator Gases to C11+

	Sample Number	2.490	2.550	2.570
	Cylinder Number	A01029	A01529	A01442
	Sample Date	09/02/15	09/02/15	09/02/15
	Sample Time	20:35	21:00	21:18
	Component	Mole%	Mole%	Mole%
H_2	Hydrogen	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000
CO_2	Carbon dioxide	0.401	0.398	0.398
N_2	Nitrogen	1.615	1.608	1.610
C_1	Methane	89.665	89.628	89.130
C_2	Ethane	4.793	4.821	4.787
C_3	Propane	1.944	1.965	1.943
iC_4	i-Butane	0.451	0.452	0.451
nC_4	n-Butane	0.456	0.461	0.453
C_5	neo-Pentane	0.006	0.006	0.006
iC_5	i-Pentane	0.219	0.220	0.213
nC_5	n-Pentane	0.137	0.138	0.134
C_6	Hexanes	0.142	0.143	0.163
	Me-Cyclo-pentane	0.024	0.024	0.032
	Benzene	0.009	0.009	0.011
	Cyclo-hexane	0.032	0.031	0.058
C_7	Heptanes	0.044	0.043	0.120
	Me-Cyclo-hexane	0.024	0.023	0.090
	Toluene	0.009	0.008	0.039
C_8	Octanes	0.017	0.015	0.173
	Ethyl-benzene	0.001	0.001	0.010
	Meta/Para-xylene	0.002	0.001	0.028
	Ortho-xylene	0.001	0.000	0.008
C_9	Nonanes	0.006	0.004	0.108
	1,2,4-TMB	0.000	0.000	0.000
C_{10}	Decanes	0.002	0.001	0.032
C_{11+}	Undecanes plus	0.000	0.000	0.003
	Totals :	100.000	100.000	100.000
	Sample Properties			
	Real Relative Density (Air=1.000 at 14.73 psia and 60°F)	0.636	0.637	0.655
	Molecular Weight (g mol-1)	18.4	18.4	18.9

Note: 0.000 means < 0.0005

Summary of Separator Liquid Sample Transfers and Sample Volumes

Cylinder Number	Core Lab Transfer Cylinder	Transfer Conditions :- Pressure (psig)	Temperature (°F)	Transfer Sample Volume (cm3)	*Sample Volume Remaining (cm3)	Date Separator Liquid Sample Cylinder Returned
4504-EA		Sample not transferred				18th June 2015
4516-EA		Sample not transferred				18th June 2015
5205-EA		Sample not transferred				18th June 2015
4337-EA		Sample not transferred				18th June 2015
5727-EA		Sample not transferred				18th June 2015
4603-EA		Sample not transferred				18th June 2015
5684-EA		Sample not transferred				18th June 2015
4479-EA		Sample not transferred				18th June 2015
14110-QA	830-157	2000	200	550	550	18th June 2015
4654-EA		Sample not transferred				18th June 2015
5216-EA	1829-02	2000	200	550	550	18th June 2015
5598-EA	1574-04	2000	200	550	550	18th June 2015

*Approximate sample volume remaining on 25th June 2015.

Summary of Separator Gas Sample Transfers

Cylinder Number	Core Lab Transfer Cylinder	Final Cylinder :- Pressure (psig)	Final Cylinder :- Temperature (°F)	Transfer Sample Cylinder Volume (cm3)	Date Separator Gas Sample Cylinder Returned
A01034		Sample not transferred			18th June 2015
A01439		Sample not transferred			18th June 2015
A01055		Sample not transferred			18th June 2015
1768		Sample not transferred			18th June 2015
A01453		Sample not transferred			18th June 2015
A01416		Sample not transferred			18th June 2015
A01422		Sample not transferred			18th June 2015
A01443		Sample not transferred			18th June 2015
A01429	A2406	128	60	20000	18th June 2015
A01029		Sample not transferred			18th June 2015
A01529	A02410	118	60	20000	18th June 2015
A01442	A03185	118	60	20000	18th June 2015

Section F - Calculated Wellstream Composition
Main Flow Period 2 Separator Samples A01529 and 5216-EA

Compositional Analysis of Main Flow Period 2 Separator Liquid Sample 5216-EA to C36+

	Component	Flashed Liquid		Flashed Gas	Separator Liquid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon Dioxide	0.000	0.000	0.568	0.043	0.007
N_2	Nitrogen	0.000	0.000	0.383	0.029	0.003
C_1	Methane	0.076	0.004	61.789	4.794	0.272
C_2	Ethane	0.172	0.017	12.940	1.148	0.122
C_3	Propane	0.482	0.070	10.545	1.251	0.195
iC_4	i-Butane	0.361	0.069	3.273	0.583	0.120
nC_4	n-Butane	0.606	0.116	3.534	0.830	0.171
C_5	Neo-Pentane	0.000	0.000	0.046	0.004	0.001
iC_5	i-Pentane	0.703	0.167	1.944	0.798	0.204
nC_5	n-Pentane	0.627	0.149	1.270	0.676	0.173
C_6	Hexanes	1.586	0.450	1.463	1.575	0.480
	M-C-Pentane	0.559	0.155	0.262	0.537	0.160
	Benzene	0.249	0.064	0.102	0.238	0.066
	Cyclohexane	0.631	0.175	0.368	0.611	0.182
C_7	Heptanes	2.082	0.687	0.536	1.964	0.696
	M-C-Hexane	1.497	0.484	0.320	1.407	0.489
	Toluene	0.748	0.227	0.122	0.700	0.228
C_8	Octanes	3.060	1.151	0.281	2.847	1.151
	E-Benzene	0.249	0.087	0.013	0.231	0.087
	M/P-Xylene	0.984	0.344	0.038	0.912	0.342
	O-Xylene	0.306	0.107	0.010	0.283	0.106
C_9	Nonanes	2.872	1.213	0.129	2.662	1.208
	1,2,4-TMB	0.432	0.171	0.000	0.399	0.170
C_{10}	Decanes	3.884	1.820	0.057	3.592	1.808
C_{11}	Undecanes	3.911	1.893	0.007	3.612	1.879
C_{12}	Dodecanes	4.050	2.147	0.000	3.740	2.130
C_{13}	Tridecanes	4.493	2.589	0.000	4.149	2.569
C_{14}	Tetradecanes	4.445	2.781	0.000	4.105	2.760
C_{15}	Pentadecanes	5.045	3.422	0.000	4.659	3.396
C_{16}	Hexadecanes	4.264	3.117	0.000	3.938	3.093
C_{17}	Heptdecanes	3.727	2.909	0.000	3.442	2.887
C_{18}	Octadecanes	3.968	3.280	0.000	3.665	3.255
C_{19}	Nonadecanes	3.725	3.226	0.000	3.440	3.201
C_{20}	Eicosanes	3.338	3.023	0.000	3.083	3.000
C_{21}	Heneicosanes	3.028	2.902	0.000	2.797	2.880
C_{22}	Docosanes	2.825	2.837	0.000	2.609	2.815
C_{23}	Tricosanes	2.489	2.606	0.000	2.298	2.586
C_{24}	Tetracosanes	2.345	2.556	0.000	2.166	2.536
C_{25}	Pentacosanes	2.065	2.346	0.000	1.907	2.328
C_{26}	Hexacosanes	1.898	2.244	0.000	1.753	2.227
C_{27}	Heptacosanes	1.705	2.100	0.000	1.575	2.084
C_{28}	Octacosanes	1.637	2.092	0.000	1.512	2.076
C_{29}	Nonacosanes	1.602	2.121	0.000	1.480	2.105
C_{30}	Triacontanes	1.513	2.072	0.000	1.397	2.056
C_{31}	Hentriacontanes	1.397	1.978	0.000	1.290	1.963
C_{32}	Dotriacontanes	1.248	1.824	0.000	1.152	1.810
C_{33}	Tritriacontanes	1.028	1.551	0.000	0.950	1.539
C_{34}	Tetratriacontanes	0.918	1.427	0.000	0.848	1.416
C_{35}	Pentatriacontanes	0.806	1.290	0.000	0.744	1.280
C_{36+}	Hexatriacontanes Plus	10.364	31.940	0.000	9.575	31.688
	Totals:	100.000	100.000	100.000	100.000	100.000

Note: 0.000 means < 0.0005.

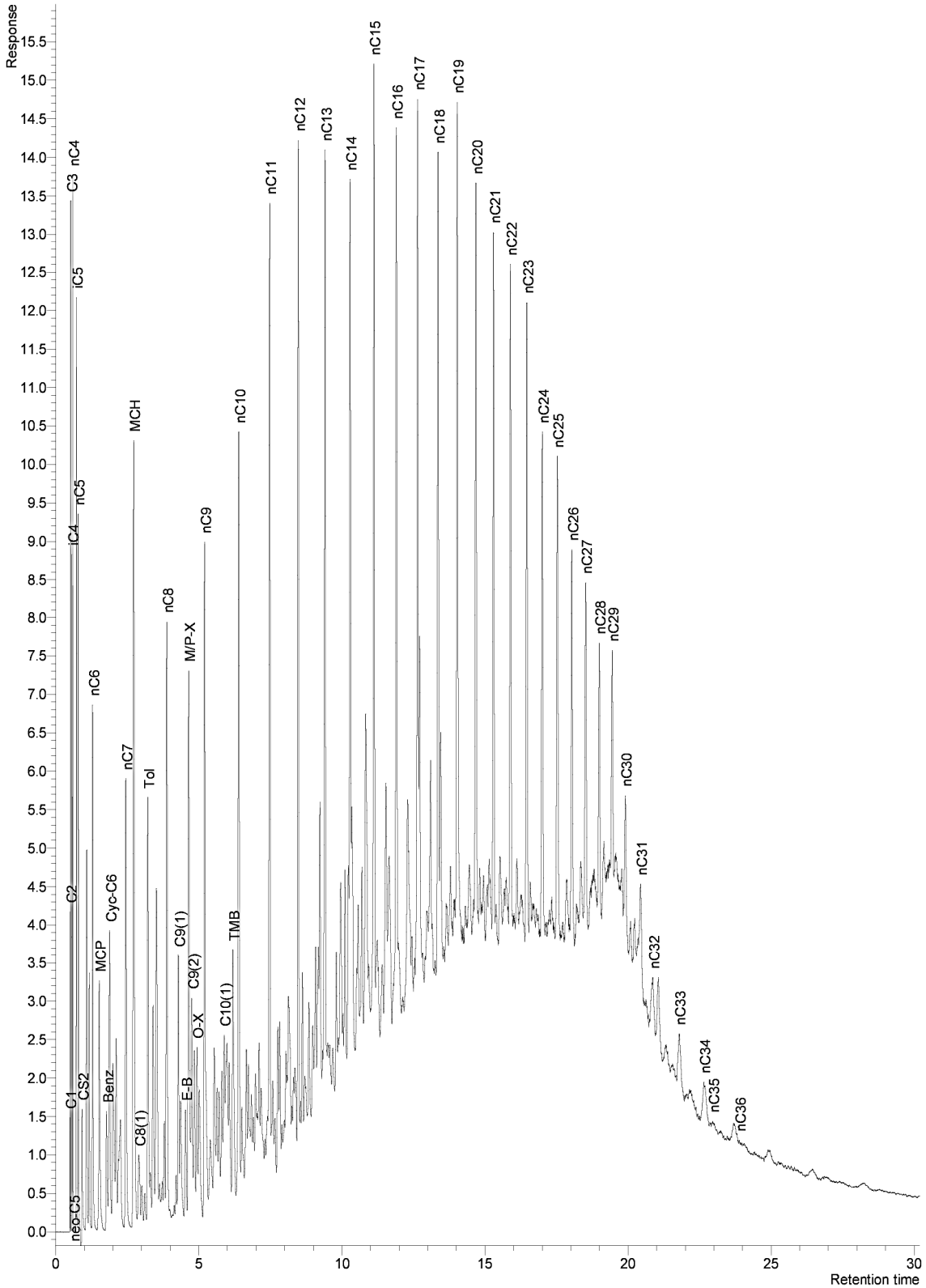
Compositional Analysis of Main Flow Period 2 Separator Liquid Sample 5216-EA to C36+

Calculated Residue Properties	Flashed Liquid	Flashed Gas	Separator Liquid
C_{7+}			
Mole%	95.387	2.245	88.269
Molecular Weight (g mol-1)	315	98.2	315
Density at 60°F (g cm-3)	0.9019	0.7534	0.9018
C_{11+}			
Mole%	77.834	0.007	71.886
Molecular Weight (g mol-1)	360	147	360
Density at 60°F (g cm-3)	0.9161	0.7890	0.9161
C_{20+}			
Mole%	40.206		37.136
Molecular Weight (g mol-1)	505		505
Density at 60°F (g cm-3)	0.9532		0.9533
C_{36+}			
Mole %	10.364		9.575
Molecular Weight (g mol-1)	936		936
Density at 60°F (g cm-3)	1.0359		1.0359

Calculated Whole Sample Properties	Flashed Liquid	Flashed Gas	Separator Liquid
Average mole weight (g mol-1)	304	28.6	283
Measured density at 60°F (g cm-3)	0.8975		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.994	
Entrained water content (wt%)	0.03		

Note: Data from atmospheric flash performed at 120°F.

Chromatographic Profile of Flashed Liquid - Main Flow Period 2 Separator Liquid Sample 5216-EA



Gas-Oil Ratio Calculations - Main Flow Period 2 Separator Samples A01529 and 5216-EA

Field Data

Standard Conditions	Pressure base	14.73	psia
	Temperature base	60	°F
Separator Conditions	Separator pressure	172	psig
	Separator temperature	81	°F
Gas Flow Data	Gas gravity factor, Fgf	n/a *	
	Supercompressibility factor, Fpvf	n/a *	
	Gas flow rate, Qf	100000.0	scf / day
Liquid Flow Data	Shrinkage factor	0.000	
	B S & W	0.0	%
	Separator Oil Flow Rate	971.0	bbl / day at 172 psig and 81°F

* - data unavailable, assumed to be equal to laboratory measured value.

Laboratory Adjusted Data

Standard Conditions	Pressure base		14.73	psia
	Temperature base		60	°F
Gas Flow Data	Gas gravity factor, Fgl		1.2533	
	Supercompressibility factor, Fpvl		1.0149	
	Gas flow rate,	=	Qf x (Fgl/Fgf) x (Fpvl/Fpvf) x Fpb x Ftb	
		=	100000.0	scf / day
Liquid Flow Data	Shrinkage factor	=	0.000	
	Separator Oil Flow Rate	=	971.0	bbl / day at 172 psig and 81°F

	Gas-Oil Ratio (Adjusted)	=	100000.0 / 971.0	scf / bbl at 172 psig and 81°F
		=	103.0	scf / bbl at 172 psig and 81°F

Fg is the gas gravity factor, the reciprocal of the square root of the gas gravity.
Fpv is the supercompressibility factor, the reciprocal of the square root of the deviation factor.
Qf is the gas flow rate.
Fpb is (field pressure base) / (report pressure base).
Ftb is (report temperature base) / (field temperature base), temperatures in Rankine.

Compositional Analysis of Main Flow Period 2 Separator Samples A01529 and 5216-EA and Calculated Wellstream to C36+

	Component	Separator Liquid		Separator Gas	Wellstream	
		Mole %	Weight %	Mole %	Mole %	Weight %
H_2	Hydrogen	0.000	0.000	0.000	0.000	0.000
H_2S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO_2	Carbon dioxide	0.043	0.007	0.398	0.115	0.022
N_2	Nitrogen	0.029	0.003	1.608	0.345	0.042
C_1	Methane	4.794	0.272	89.628	21.714	1.515
C_2	Ethane	1.148	0.122	4.821	1.880	0.246
C_3	Propane	1.251	0.195	1.965	1.392	0.267
iC_4	i-Butane	0.583	0.120	0.452	0.557	0.141
nC_4	n-Butane	0.830	0.171	0.461	0.758	0.192
C_5	neo-Pentane	0.004	0.001	0.006	0.004	0.001
iC_5	i-Pentane	0.798	0.204	0.220	0.684	0.215
nC_5	n-Pentane	0.676	0.173	0.138	0.570	0.179
C_6	Hexanes	1.575	0.480	0.143	1.289	0.483
	Me-Cyclo-pentane	0.537	0.160	0.024	0.435	0.159
	Benzene	0.238	0.066	0.009	0.193	0.066
	Cyclo-hexane	0.611	0.182	0.031	0.496	0.181
C_7	Heptanes	1.964	0.696	0.043	1.580	0.689
	Me-Cyclo-hexane	1.407	0.489	0.023	1.131	0.483
	Toluene	0.700	0.228	0.008	0.561	0.225
C_8	Octanes	2.847	1.151	0.015	2.283	1.134
	Ethyl-benzene	0.231	0.087	0.001	0.186	0.086
	Meta/Para-xylene	0.912	0.342	0.001	0.729	0.337
	Ortho-xylene	0.283	0.106	0.000	0.226	0.104
C_9	Nonanes	2.662	1.208	0.004	2.132	1.189
	Tri-Me-benzene	0.399	0.170	0.000	0.320	0.167
C_{10}	Decanes	3.592	1.808	0.001	2.875	1.779
C_{11}	Undecanes	3.612	1.879	0.000	2.892	1.849
C_{12}	Dodecanes	3.740	2.130	0.000	2.993	2.096
C_{13}	Tridecanes	4.149	2.569	0.000	3.321	2.528
C_{14}	Tetradecanes	4.105	2.760	0.000	3.287	2.716
C_{15}	Pentadecanes	4.659	3.396	0.000	3.730	3.342
C_{16}	Hexadecanes	3.938	3.093	0.000	3.152	3.044
C_{17}	Heptadecanes	3.442	2.887	0.000	2.756	2.841
C_{18}	Octadecanes	3.665	3.255	0.000	2.934	3.203
C_{19}	Nonadecanes	3.440	3.201	0.000	2.754	3.150
C_{20}	Eicosanes	3.083	3.000	0.000	2.468	2.952
C_{21}	Heneicosanes	2.797	2.880	0.000	2.239	2.834
C_{22}	Docosanes	2.609	2.815	0.000	2.088	2.770
C_{23}	Tricosanes	2.298	2.586	0.000	1.840	2.545
C_{24}	Tetracosanes	2.166	2.536	0.000	1.733	2.496
C_{25}	Pentacosanes	1.907	2.328	0.000	1.527	2.291
C_{26}	Hexacosanes	1.753	2.227	0.000	1.404	2.191
C_{27}	Heptacosanes	1.575	2.084	0.000	1.261	2.051
C_{28}	Octacosanes	1.512	2.076	0.000	1.211	2.043
C_{29}	Nonacosanes	1.480	2.105	0.000	1.185	2.071
C_{30}	Triacontanes	1.397	2.056	0.000	1.118	2.023
C_{31}	Hentriacontanes	1.290	1.963	0.000	1.033	1.932
C_{32}	Dotriacontanes	1.152	1.810	0.000	0.922	1.781
C_{33}	Tritriacontanes	0.950	1.539	0.000	0.760	1.514
C_{34}	Tetratriacontanes	0.848	1.416	0.000	0.679	1.393
C_{35}	Pentatriacontanes	0.744	1.280	0.000	0.596	1.260
C_{36+}	Hexatriacontanes plus	9.575	31.688	0.000	7.662	31.182
	Totals:	100.000	100.000	100.000	100.000	100.000

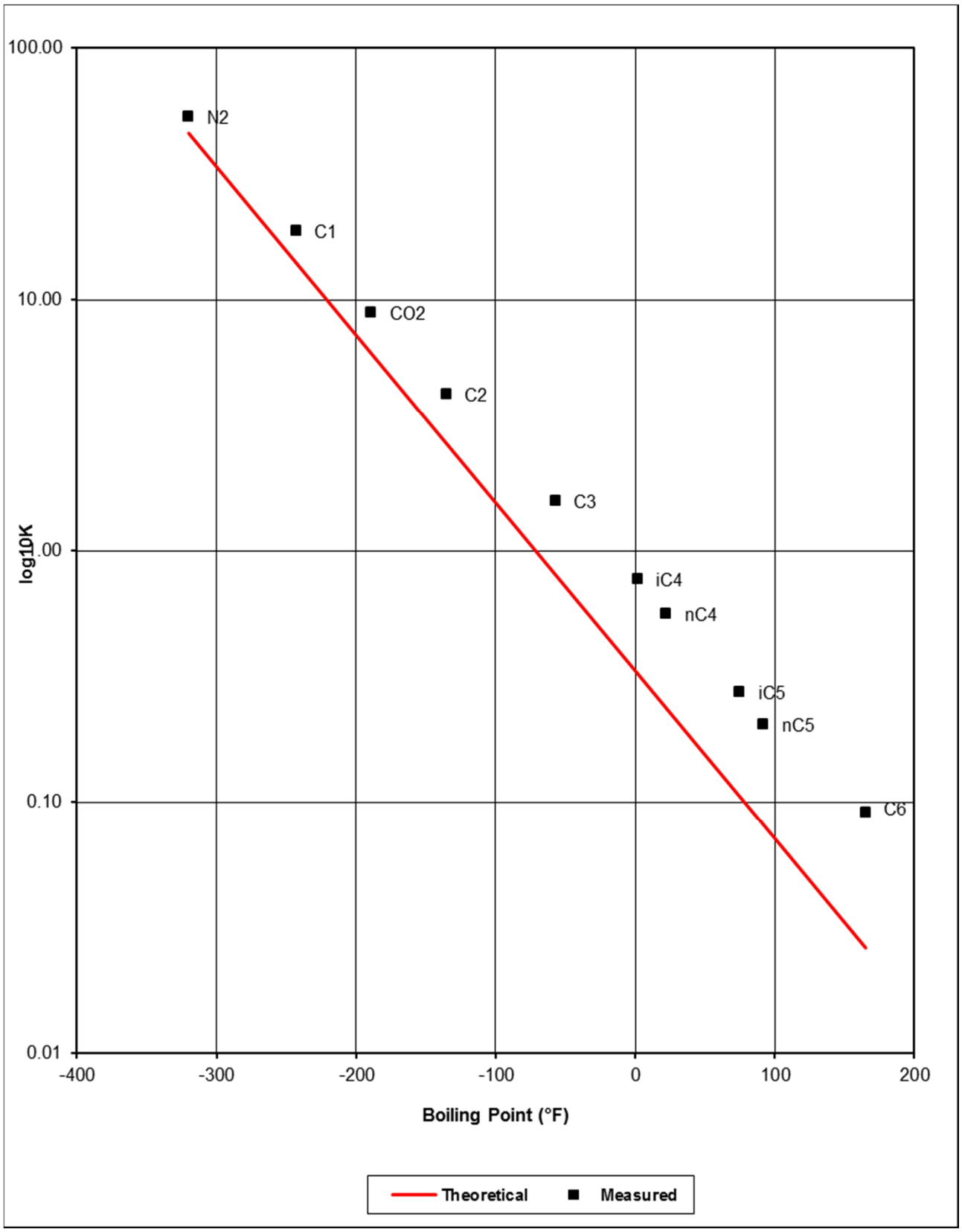
Note: 0.000 means < 0.0005.

Compositional Analysis of Main Flow Period 2 Separator Samples A01529 and 5216-EA and Calculated Wellstream to C36+

Calculated Properties	Separator Liquid	Separator Gas	Wellstream
C_{7+}			
Mole%	88.269	0.160	70.692
Molecular Weight (g mol-1)	315	96.6	315
Density at 60°F (g cm-3)	0.9018	0.7450	0.9018
C_{11+}			
Mole%	71.886		57.545
Molecular Weight (g mol-1)	360		360
Density at 60°F (g cm-3)	0.9161		0.9161
C_{20+}			
Mole%	37.136		29.726
Molecular Weight (g mol-1)	505		505
Density at 60°F (g cm-3)	0.9533		0.9532
C_{36+}			
Mole %	9.575		7.662
Molecular Weight (g mol-1)	936		936
Density at 60°F (g cm-3)	1.0359		1.0359

Calculated Whole Sample Properties	Separator Liquid	Separator Gas	Wellstream
Average mole weight (g mol-1)	283	18.4	230
Density at separator conditions (g cm-3 at 172 psig and 81°F)	0.8849		
Real relative density (Air = 1.000 at 14.73 psia and 60°F)		0.637	

Graph of log(Equilibrium k) v Boiling Point Temperature
Main Flow Period 2 Separator Samples A01529 and 5216-EA



Note: Theoretical k Values ref. GPSA Engineering Data Book, Vol I, Section 25, "Equilibrium Ratio (K) Data",1987.

Section G - Crude Oil Inspection Data

Crude Oil Inspection Data - Atmospheric Oil Sample 2.53

Analysis	Method	Result	Units
Water content	IP 356	0.03	% m/m
Sediment	ASTM D4807	0.004	% m/m
Salt	UOP 22	21	mg/l NaCl
Molecular weight	Cryoscope	304	g mol -1
Density at 60°F	IP 365	0.8991	g cm -3
API	IP 365	25.7	°
Pour point	IP 441	9	°C
Wax appearance temperature	IP 389	29	°C
Wax	BP Modified	7.5	% m/m
Total acid number	IP 177	0.80	mg KOH/g
Sulphur	IP 336	0.37	% m/m
Mercaptans	UOP 163	< 1	ppm wt S
H_2S	UOP 163	Nil	ppm wt S
Vanadium	ICP	2	mg/kg
Nickel	ICP	8	mg/kg
Iron	ICP	< 1	mg/kg
Calcium	ICP	1	mg/kg
Total Mercury	ASTM D6722	< 1	ng/g
Viscosity at 20°C	IP 71	88.39	cSt
Viscosity at 40°C	IP 71	32.38	cSt
Viscosity at 60°C	IP 71	15.21	cSt
Asphaltenes (C5)	Core Lab	3.2	% m/m
Asphaltenes (C7)	IP 143	0.55	% m/m
Saturates	Iatroscan	45.9	% m/m
Aromatics	Iatroscan	30.8	% m/m
Resins	Iatroscan	17.3	% m/m
Asphaltenes	Iatroscan	6.0	% m/m

Compositional Analysis of Atmospheric Oil 2.53 to C36+

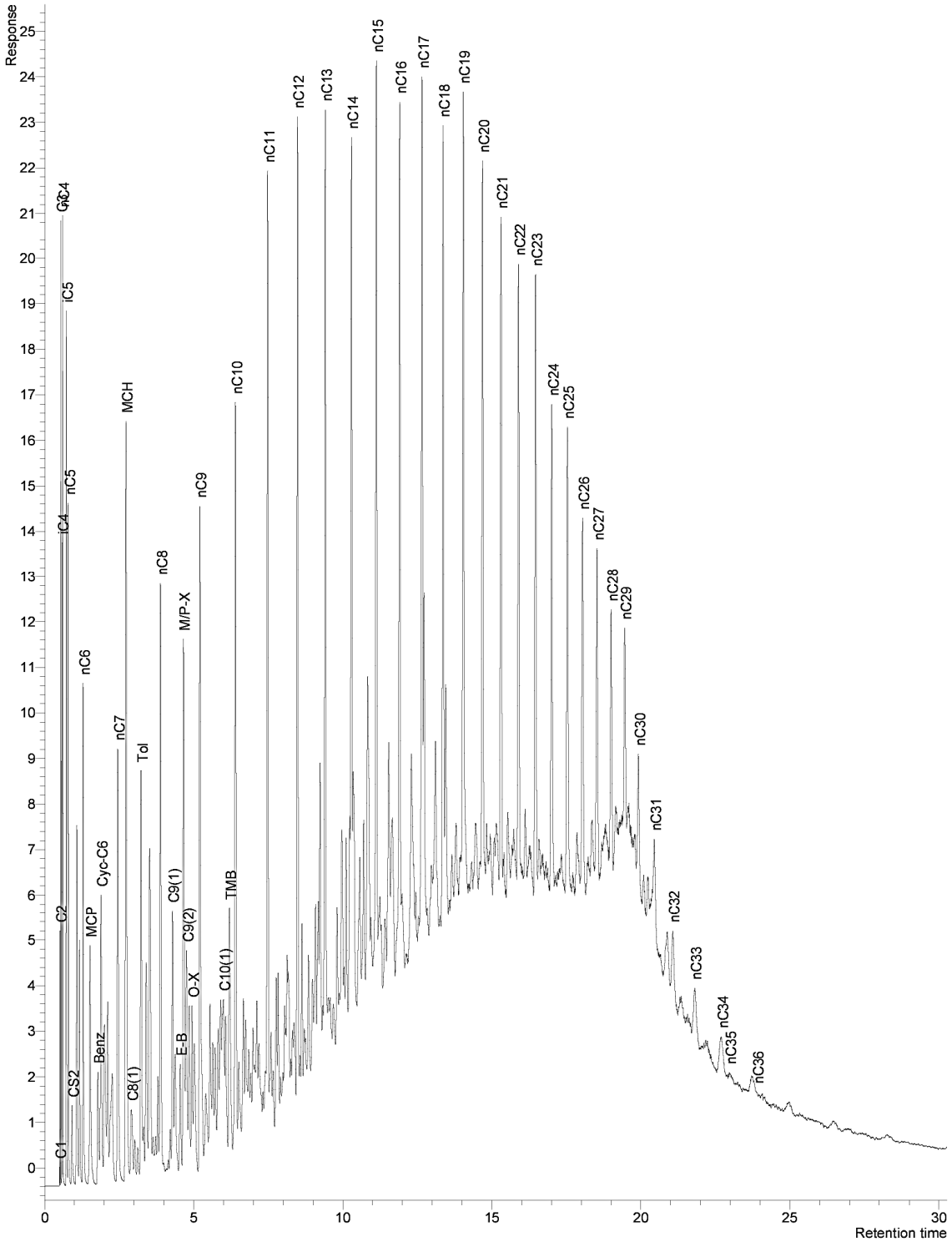
	Component	Mole %	Weight %
H_2	Hydrogen	0.000	0.000
H_2S	Hydrogen Sulphide	0.000	0.000
CO_2	Carbon Dioxide	0.000	0.000
N_2	Nitrogen	0.000	0.000
C_1	Methane	0.019	0.001
C_2	Ethane	0.142	0.014
C_3	Propane	0.469	0.068
iC_4	i-Butane	0.340	0.065
nC_4	n-Butane	0.570	0.109
C_5	Neo-Pentane	0.000	0.000
iC_5	i-Pentane	0.675	0.160
nC_5	n-Pentane	0.586	0.139
C_6	Hexanes	1.535	0.435
	M-C-Pentane	0.539	0.149
	Benzene	0.230	0.059
	Cyclohexane	0.654	0.181
C_7	Heptanes	2.007	0.661
	M-C-Hexane	1.571	0.507
	Toluene	0.756	0.229
C_8	Octanes	3.014	1.132
	E-Benzene	0.246	0.086
	M/P-Xylene	0.974	0.340
	O-Xylene	0.301	0.105
C_9	Nonanes	2.794	1.178
	1,2,4-TMB	0.450	0.178
C_{10}	Decanes	3.837	1.795
C_{11}	Undecanes	3.925	1.897
C_{12}	Dodecanes	4.062	2.150
C_{13}	Tridecanes	4.509	2.594
C_{14}	Tetradecanes	4.442	2.775
C_{15}	Pentadecanes	5.059	3.426
C_{16}	Hexadecanes	4.317	3.151
C_{17}	Heptadecanes	3.912	3.048
C_{18}	Octadecanes	3.983	3.287
C_{19}	Nonadecanes	3.800	3.286
C_{20}	Eicosanes	3.325	3.006
C_{21}	Heneicosanes	3.029	2.898
C_{22}	Docosanes	2.826	2.834
C_{23}	Tricosanes	2.504	2.618
C_{24}	Tetracosanes	2.286	2.488
C_{25}	Pentacosanes	2.097	2.378
C_{26}	Hexacosanes	1.907	2.251
C_{27}	Heptacosanes	1.682	2.068
C_{28}	Octacosanes	1.673	2.134
C_{29}	Nonacosanes	1.672	2.210
C_{30}	Triacontanes	1.504	2.057
C_{31}	Hentriacontanes	1.432	2.025
C_{32}	Dotriacontanes	1.226	1.790
C_{33}	Tritriacontanes	1.076	1.620
C_{34}	Tetratriacontanes	0.935	1.451
C_{35}	Pentatriacontanes	0.835	1.334
C_{36+}	Hexatriacontanes plus	10.273	31.633
	Totals :	100.000	100.000

Note: 0.000 means less than 0.0005.

Compositional Analysis of Atmospheric Oil 2.53 to C36+

Calculated Residue Properties	
C_{7+}	
Mole%	95.664
Molecular Weight (g mol-1)	315
Density at 60°F (g cm-3)	0.9033
C_{11+}	
Mole%	78.291
Molecular Weight (g mol-1)	359
Density at 60°F (g cm-3)	0.9173
C_{20+}	
Mole%	40.282
Molecular Weight (g mol-1)	504
Density at 60°F (g cm-3)	0.9554
C_{36+}	
Mole%	10.273
Molecular Weight (g mol-1)	937
Density at 60°F (g cm-3)	1.0425
Calculated Whole Sample Properties	
Average mole weight (g mol-1)	304
Density at 60°F (g cm-3) [Measured]	0.8991
API (°)	25.7

Chromatographic Profile of Atmospheric Oil 2.53



Section H - Appendix

Data Used in Gas Compositional Calculations

Component		Mole Weight (g mol-1)	Density (g cm-3 at 60°F)	Component		Mole Weight (g mol-1)	Density (g cm-3 at 60°F)
Hydrogen	*	2.016	N/A	33DMC5	*	100.20	0.6954
Oxygen/(Argon)	**	31.999	1.1410	Cyclohexane	*	84.16	0.7827
Nitrogen (Corrected)	**	28.013	0.8086	2MC6/23DMC5	*	100.20	0.6917
Methane	**	16.043	0.2997	11DMCYC5/3MC6	*	99.20	0.7253
Carbon Dioxide	**	44.010	0.8172	t13DMCYC5	*	98.19	0.7528
Ethane	**	30.070	0.3558	c13DMCYC5/3EC5	*	99.20	0.7262
Hydrogen Sulphide	**	34.080	0.8006	t12DMCYC5	*	98.19	0.7554
Propane	**	44.097	0.5065	Heptanes (nC7)	*	100.20	0.6875
i-Butane	**	58.123	0.5623	22DMC6	*	114.23	0.6994
n-Butane	**	58.123	0.5834	MCYC6	*	98.19	0.7740
Neo-Pentane	*	72.15	0.5968	ECYC5	*	98.19	0.7704
i-Pentane	**	72.150	0.6238	223TMC5/24&25DMC6	*	114.23	0.7060
n-Pentane	**	72.150	0.6305	ctc124TMCYC5	*	112.21	0.7511
22DMC4	*	86.18	0.6529	ctc123TMCYC5	*	112.21	0.7574
23DMC4/CYC5	*	78.16	0.7129	Toluene	*	92.14	0.8734
2MC5	*	86.18	0.6572	Octanes (nC8)	*	114.23	0.7063
3MC5	*	86.18	0.6682	E-Benzene	*	106.17	0.8735
Hexanes (nC6)	*	86.18	0.6631	M/P-Xylene	*	106.17	0.8671
22DMC5	*	100.20	0.6814	O-Xylene	*	106.17	0.8840
M-C-Pentane	*	84.16	0.7533	Nonanes (nC9)	*	128.26	0.7212
24DMC5	*	100.20	0.6757	Decanes	***	134	0.778
223TMC4	*	100.20	0.6947	Undecanes	***	147	0.789
Benzene	*	78.11	0.8820	Dodecanes	***	161	0.800

Data Source Refs :

* ASTM Data Series Publication DS 4B (1991) - Physical Constants of Hydrocarbon and Non-Hydrocarbon Compounds.

** GPA Table of Physical Constants of Paraffin Hydrocarbons and Other Components of Natural Gas, GPA 2145-96.

*** Journal of Petroleum Technology, Nov 1978, Pages 1649-1655.
Predicting Phase Behaviour of Condensate/Crude Oil Systems Using Methane Interaction Coefficients - D.L. Katz & A. Firoozabadi.

Note :
The gas mole % compositions were calculated from the measured weight % compositions using the most detailed analysis results, involving as many of the above components as were identified. The reported component mole % compositions were then sub-grouped into the generic carbon number components.

Data Used in Liquid Compositional Calculations

Component		Mole Weight (g mol-1)	Density (g cm-3 at 60°F)	Component		Mole Weight (g mol-1)	Density (g cm-3 at 60°F)
Hydrogen	*	2.016	N/A	Undecanes	***	147	0.789
Hyd. sulphide	**	34.080	0.8006	Dodecanes	***	161	0.800
Carbon Dioxide	**	44.010	0.8172	Tridecanes	***	175	0.811
Nitrogen	**	28.013	0.8086	Tetradecanes	***	190	0.822
Methane	**	16.043	0.2997	Pentadecanes	***	206	0.832
Ethane	**	30.070	0.3558	Hexadecanes	***	222	0.839
Propane	**	44.097	0.5065	Heptadecanes	***	237	0.847
i-Butane	**	58.123	0.5623	Octadecanes	***	251	0.852
n-Butane	**	58.123	0.5834	Nonadecanes	***	263	0.857
i-Pentane	**	72.150	0.6238	Eicosanes	***	275	0.862
n-Pentane	**	72.150	0.6305	Heneicosanes	***	291	0.867
Hexanes	**	86.177	0.6634	Docosanes	***	305	0.872
Me-cyclo-pentane	*	84.16	0.7533	Tricosanes	***	318	0.877
Benzene	*	78.11	0.8820	Tetracosanes	***	331	0.881
Cyclo-hexane	*	84.16	0.7827	Pentacosanes	***	345	0.885
Heptanes	**	100.204	0.6874	Hexacosanes	***	359	0.889
Me-cyclo-hexane	*	98.19	0.7740	Heptacosanes	***	374	0.893
Toluene	*	92.14	0.8734	Octacosanes	***	388	0.896
Octanes	**	114.231	0.7061	Nonacosanes	***	402	0.899
Ethyl-benzene	*	106.17	0.8735	Triacontanes	***	416	0.902
Meta/Para-xylene	*	106.17	0.8671	Hentriacontanes	***	430	0.906
Ortho-xylene	*	106.17	0.8840	Dotriacontanes	***	444	0.909
Nonanes	**	128.258	0.7212	Tritriacontanes	***	458	0.912
1-2-4-T-M-benzene	*	120.19	0.8797	Tetratriacontanes	***	472	0.914
Decanes	**	142.285	0.7334	Pentatriacontanes	***	486	0.917

Data Source Refs :

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Note :
The residue mole weight and density values (eg heptanes plus, undecanes plus, eicosanes plus) are calculated so that the calculated average mole weights and densities correspond with the measured values. This can lead to anomalous residue mole weights and densities where the Katz and Firoozabadi values may not be suitable for the isomer groups detected.

Report prepared by	**Report approved by**
GDPR Redaction	GDPR Redaction
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